

MASTER BIP – Year 1-

Course Unit: Diversity of Study Models in Physiology

PLANT MODELS

Model Species and Techniques



Eugénie Carnero Diaz

eugenie.carnero_diaz@sorbonne-universite.fr

Muséum National d'Histoire Naturelle – UMR 7205 ISYEB

Learning Objectives



Plants : Powerful models to address questions in fundamental and applied physiology

Links with:

- Human health
- Genetics
- Bioengineering
- Stress adaptation

Model organisms



Definition

Living species, representative of a taxonomic group,
with properties that facilitate demonstration and
experimental practice.

Model organisms

Characteristics

- Known and homogeneous genetic background
- Easy to handle and obtain (manageable at the scale of laboratory, greenhouse or animal facility)
- Easy and rapid reproduction (to regularly renew the experimental organisms)
- Adaptable to the widest range of questions
- Facilities comparison of results (statistical analyses)

Course Outline

I. Introduction

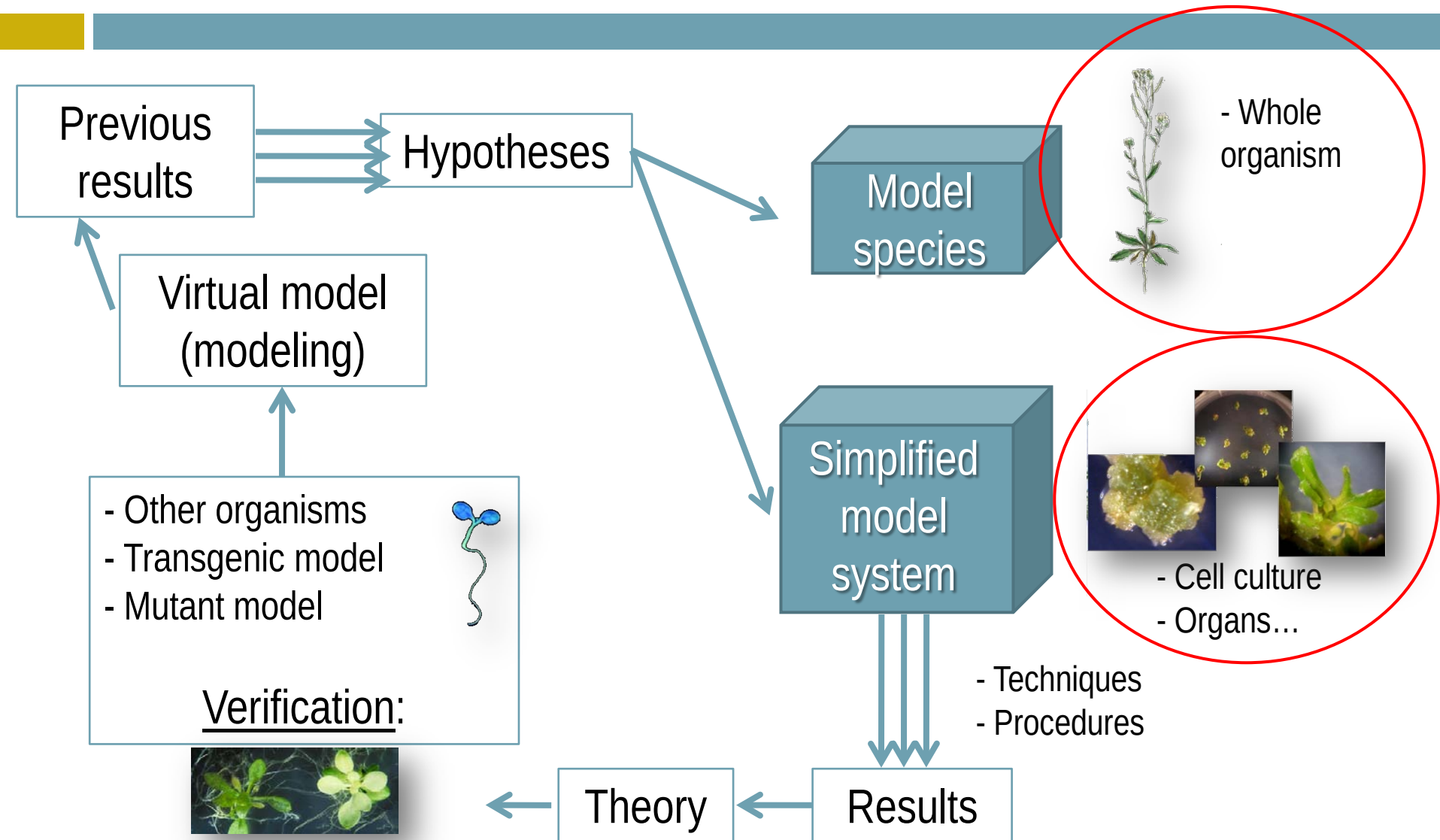
II. Totipotency and plant cell models

III. Model 1: *Arabidopsis thaliana* – “Physiological and developmental model”

IV. Model 2: *Oryza sativa* – “Agricultural and genomic model”

V. Model 3: *Populus trichocarpa* – “Woody plant model”

VI. Conclusion



I

II

III

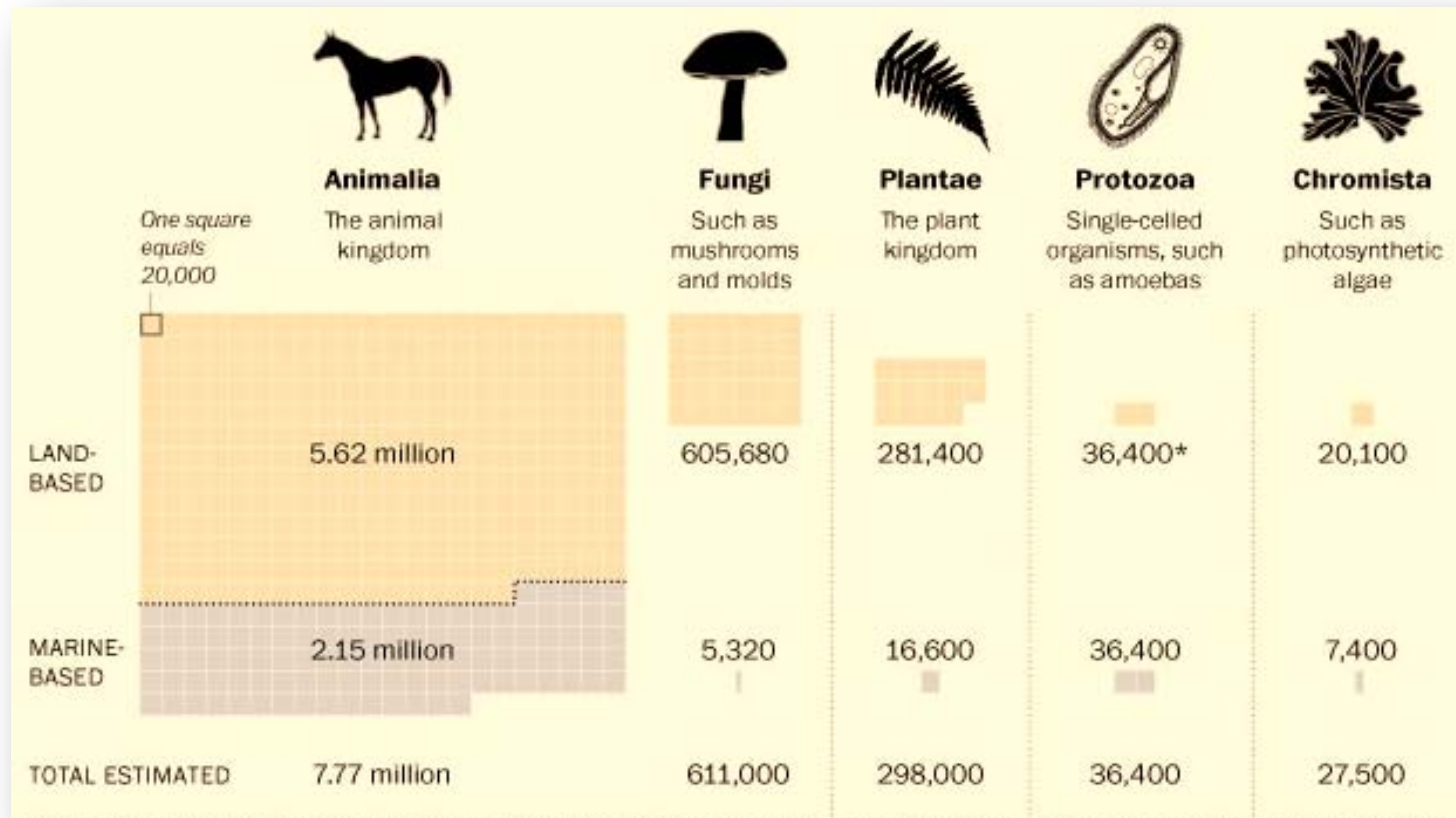
IV

V

VI



Why use model species?



Impossible to experiment on all living species



Why use model species?

→ 1980 : Identification of a small number of model species

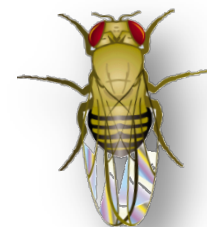
➤ Model organisms of:

- Plants → *Arabidopsis*
- Mammals → Mouse



➤ Models organisms for:

- Study of developmental biology → *Drosophila*
- Study of lignification → Poplar



I

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III

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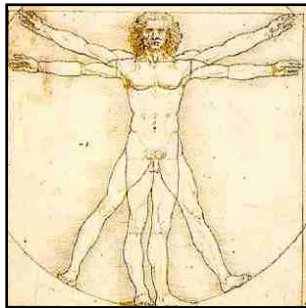
V

VI



Simplifying of the organism

➤ Each organism is complex



200 different
cell types



~ 40 different
cell types

→ If one function is disturbed, there will be consequences on other functions

→ **Cell models** are used to eliminate the various interactions between functions



Why study plants in physiology?



Mus musculus



Same tools, same questions:

- Mutagenesis
- CRISPR / genetic editing ...
- Oxidative stress
- Cell cycle
- Signaling
- Aging and regeneration ...



Arabidopsis thaliana

Understanding plants also means understanding universal mechanisms

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Plants, pioneers of modern biology



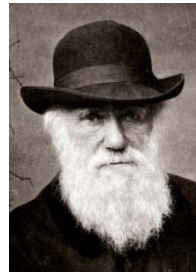
Gregor Mendel

1865

Pisum sativum



Laws of heredity

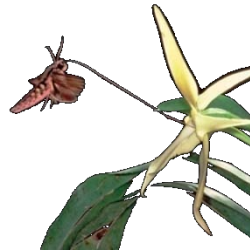


Charles Darwin

1880



Angraecum sesquipedale
Xanthopan morgani



Coevolution



Barbara McClintock

1950

Zea mays



Transposons

I

II

III

IV

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Plant models: a key advantage for biologists

Unique advantages	Health applications
• No ethical barriers • Easy totipotency and regeneration • Genome often simple and well annotated • Important genetic resources and collections	• Drug production (vinblastine, monoclonal antibodies) • Study of oxidative stress, stem cells, aging ...

From stem cells to oxidative stress, plants reveal universal principles of physiology

Course Outline

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II. Totipotency and plant cell models

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VI. Conclusion

I

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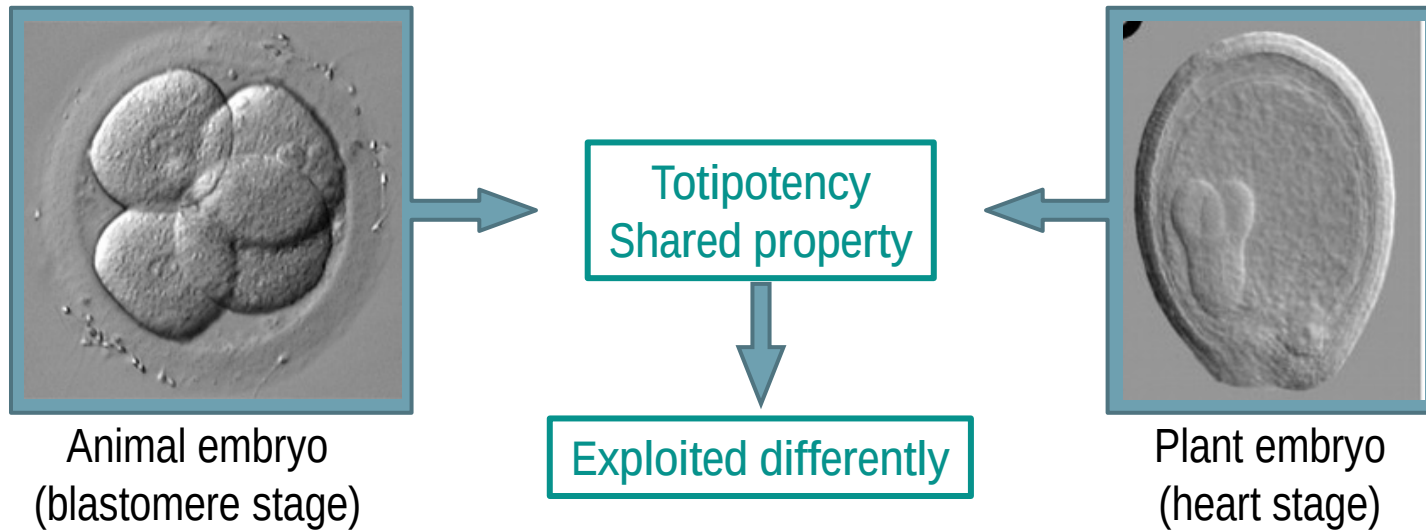
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Totipotency and plant cell models

Definition

Totipotency is the property of a cell to differentiate into any specialized cell and to regenerate a complete organism.



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III

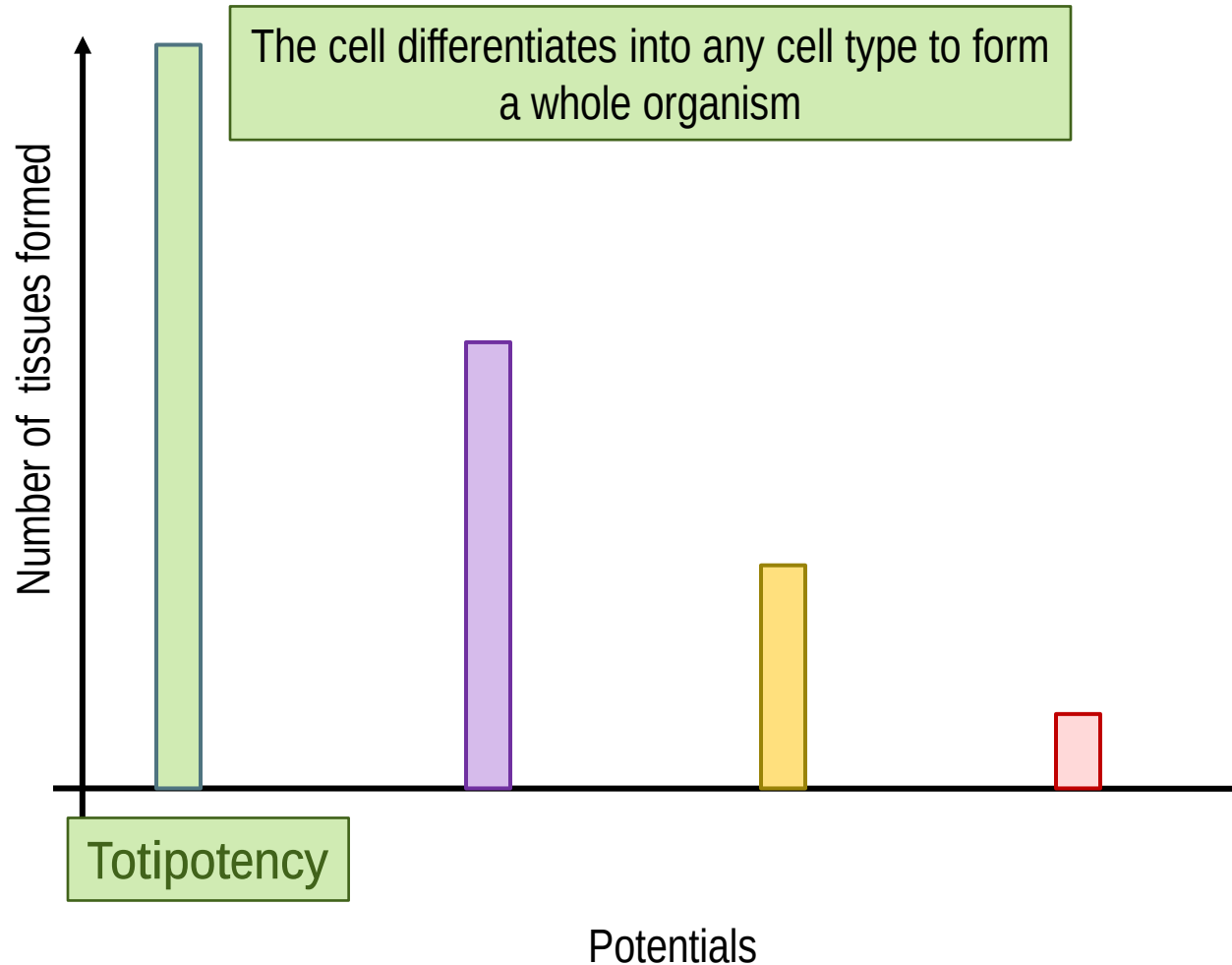
IV

V

VI



Totipotency and other cellular potentials



I

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VI



Totipotency of plants and animal cells

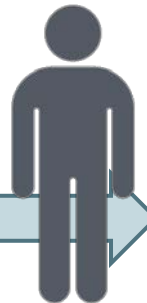
Fertilization

Divisions

Morula

Blastocyst

Gastrulation



Zygote

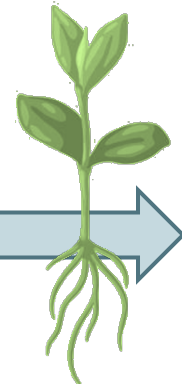
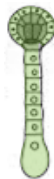
Early blastomeres
(2 and 4 cells)

Embryo
(16 to 32 cells)

Inner cell
mass

Three embryonic
layers

Totipotency



Zygote

Proembryo
(2 cells)

Globular
stage

Heart
stage

Torpedo
stage

Cell differentiation

I

II

III

IV

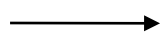
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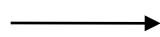


Totipotency of plants and animal cells

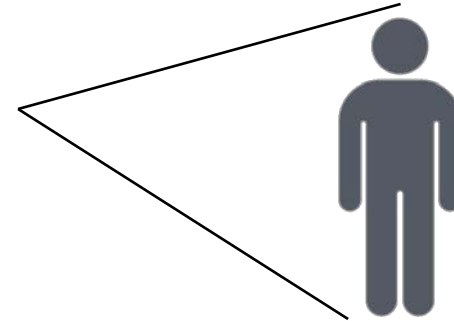
Zygote



Tissues



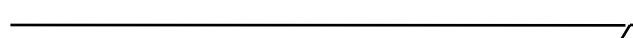
Organs



Blastomeres
(2 and 4 cells)

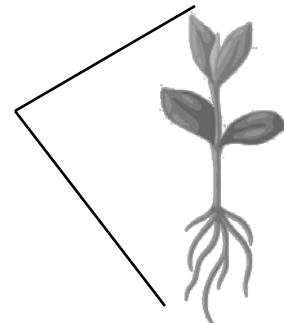
Totipotency

Zygote



Tissues

Organs



Dedifferentiation



Differentiated cells

I

II

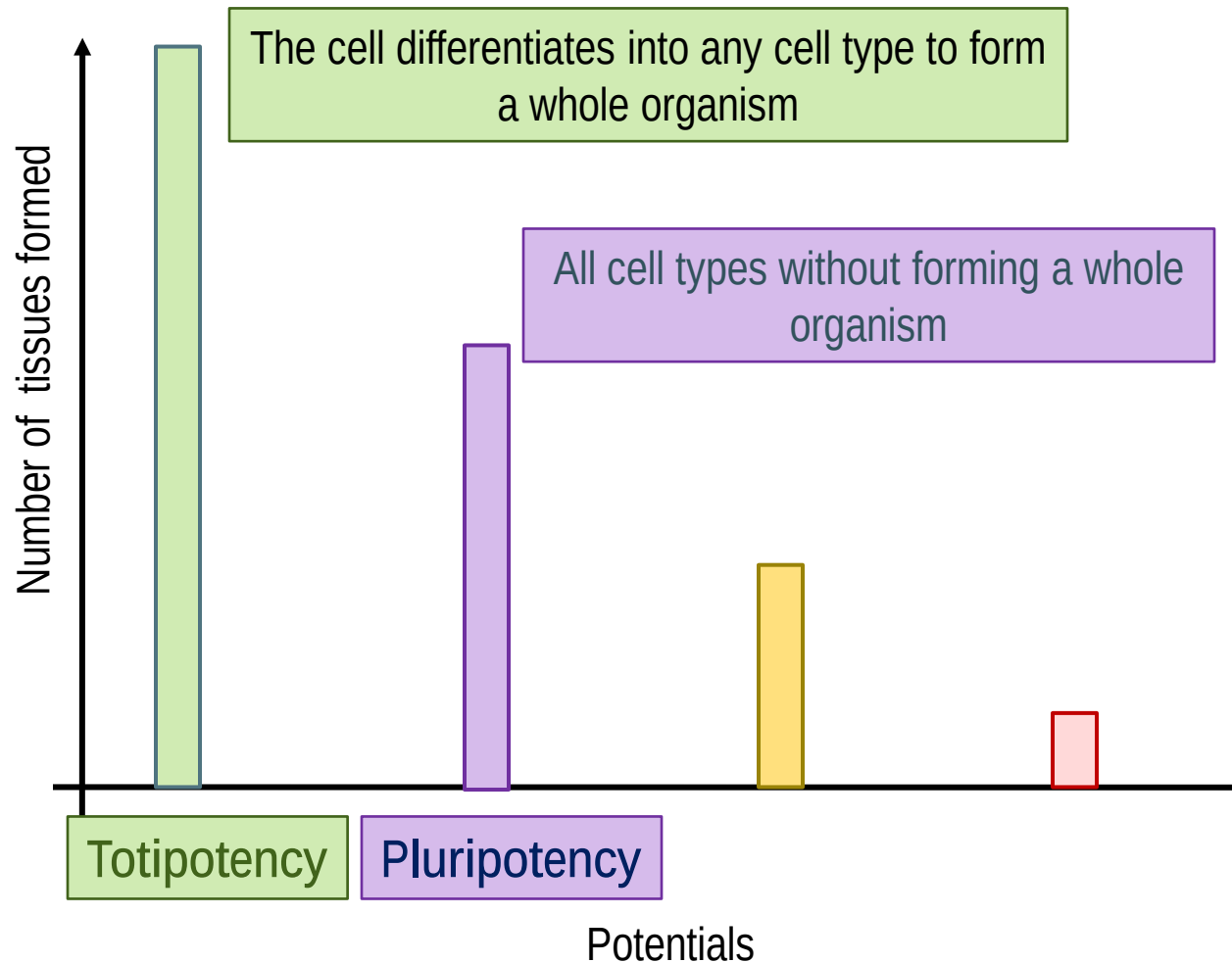
III

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Totipotency and other cellular potentials



I

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Totipotency of plants and animal cells

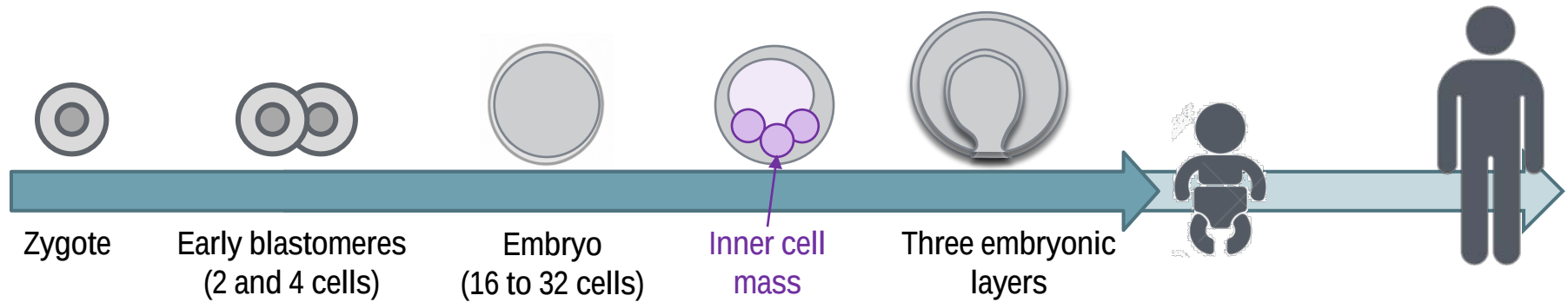
Fertilization

Divisions

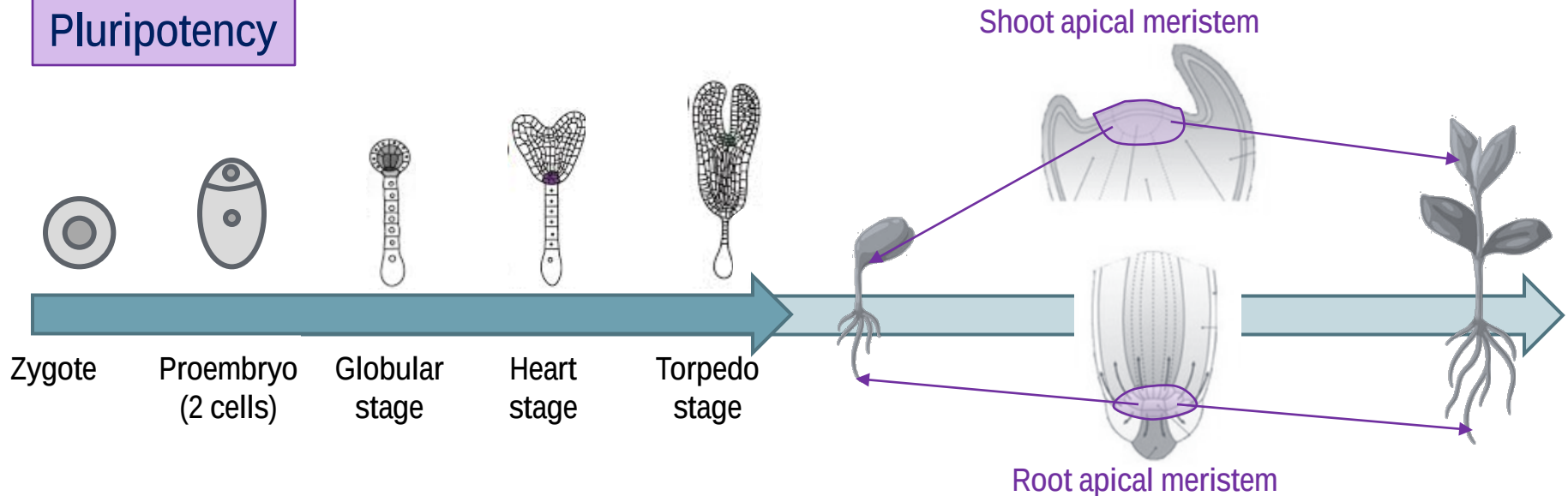
Morula

Blastocyst

Gastrulation

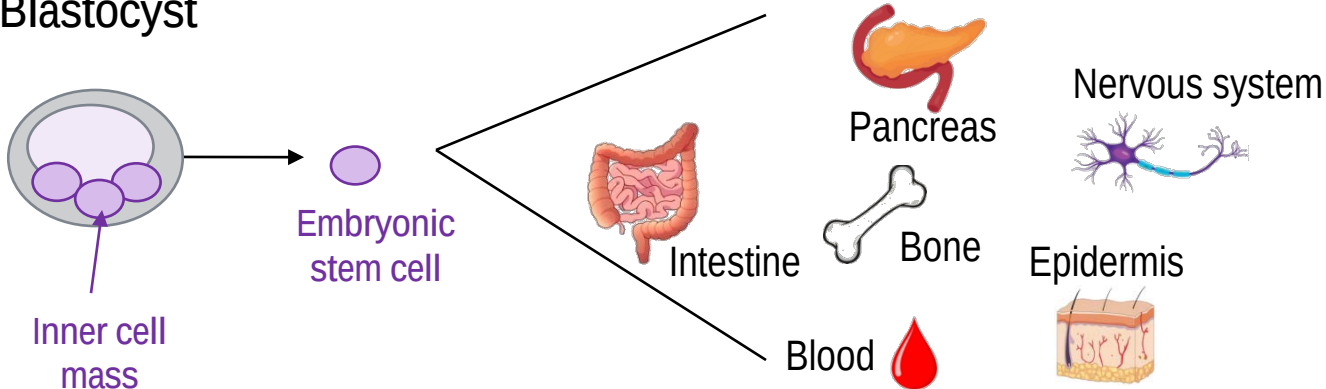


Pluripotency

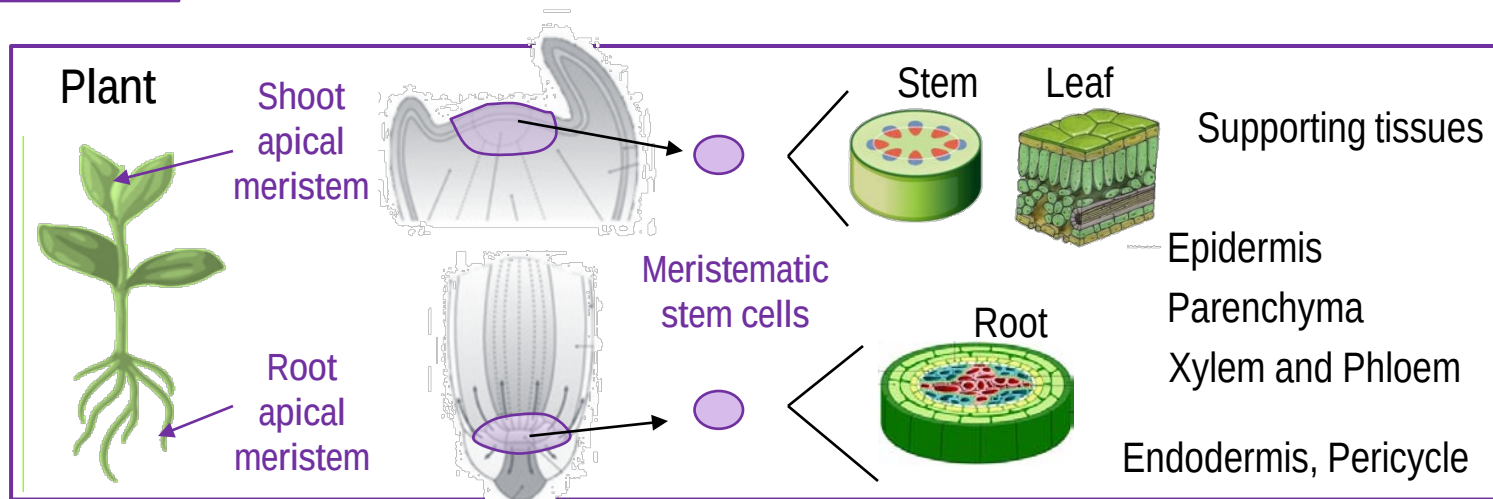


Totipotency of plants and animal cells

Blastocyst



Pluripotency



I

II

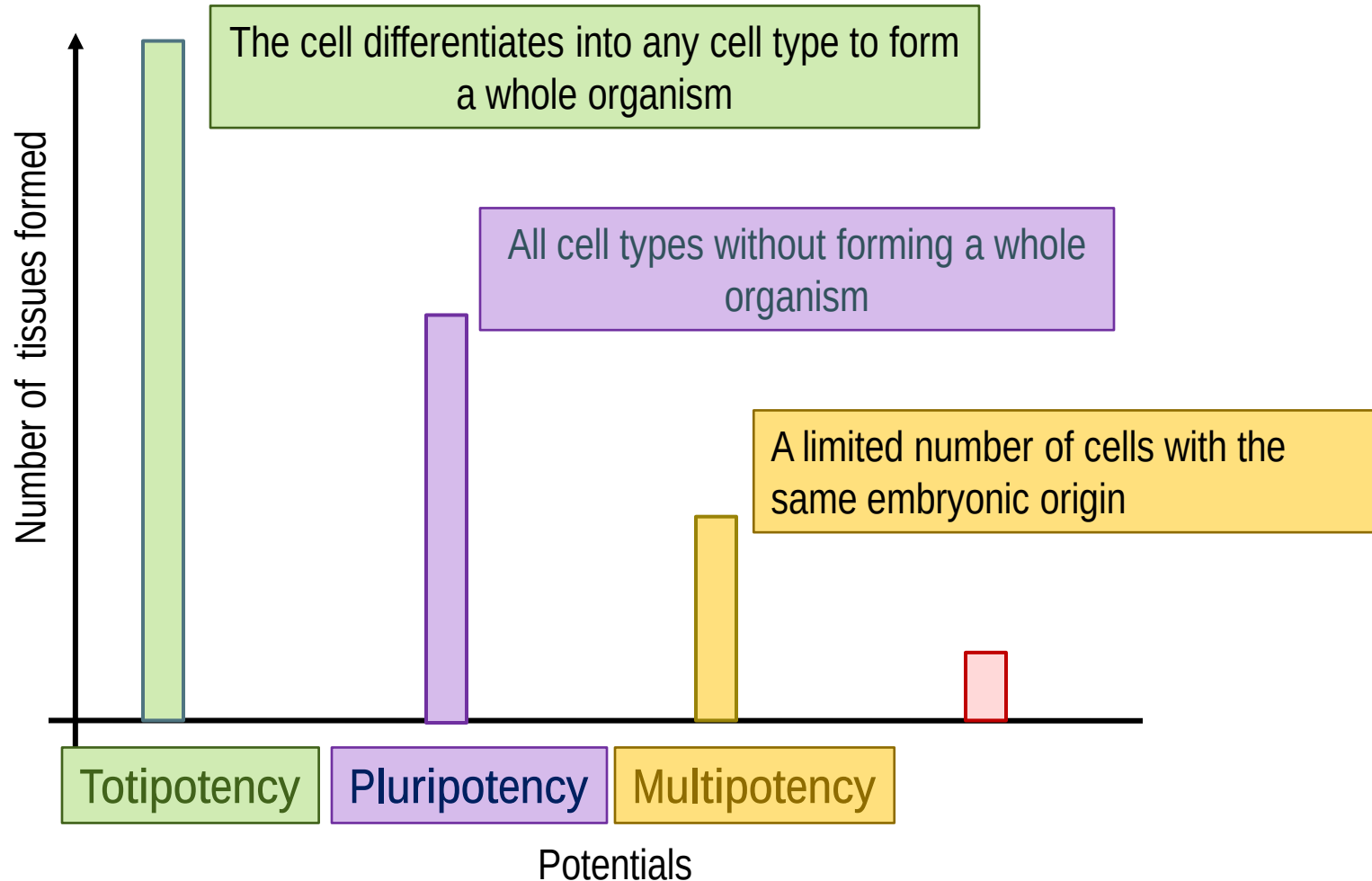
III

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Totipotency and other cellular potentials



I

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Totipotency of plants and animal cells

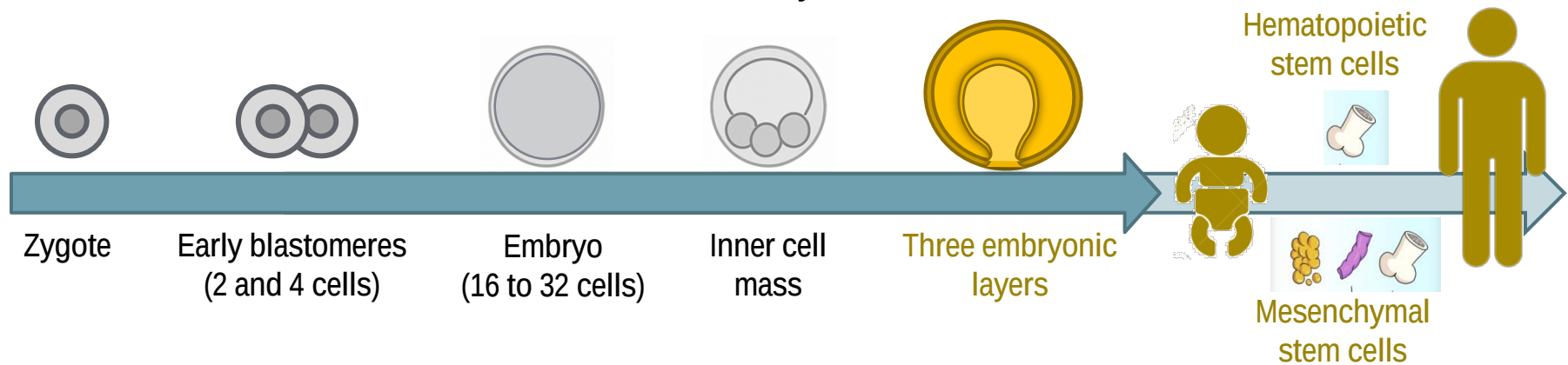
Fertilization

Divisions

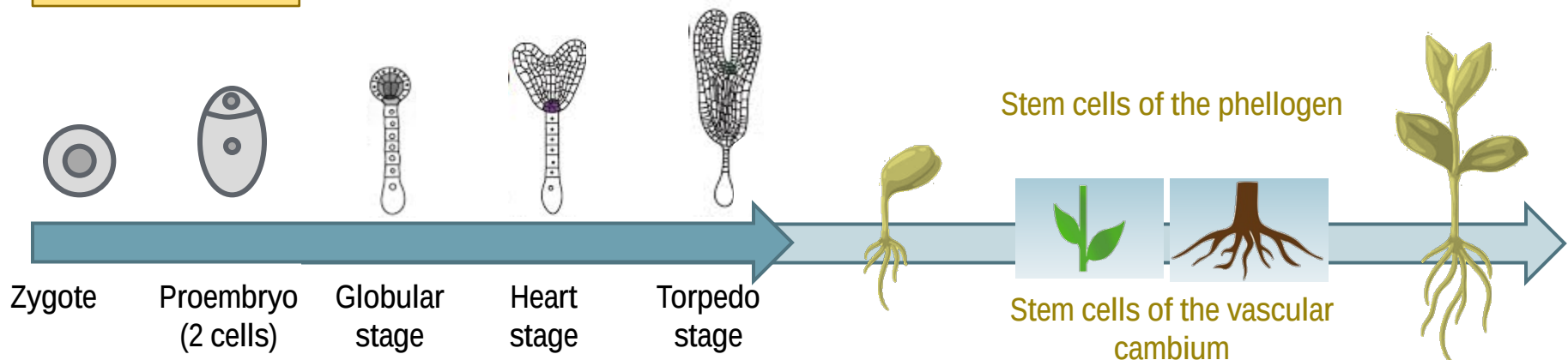
Morula

Blastocyst

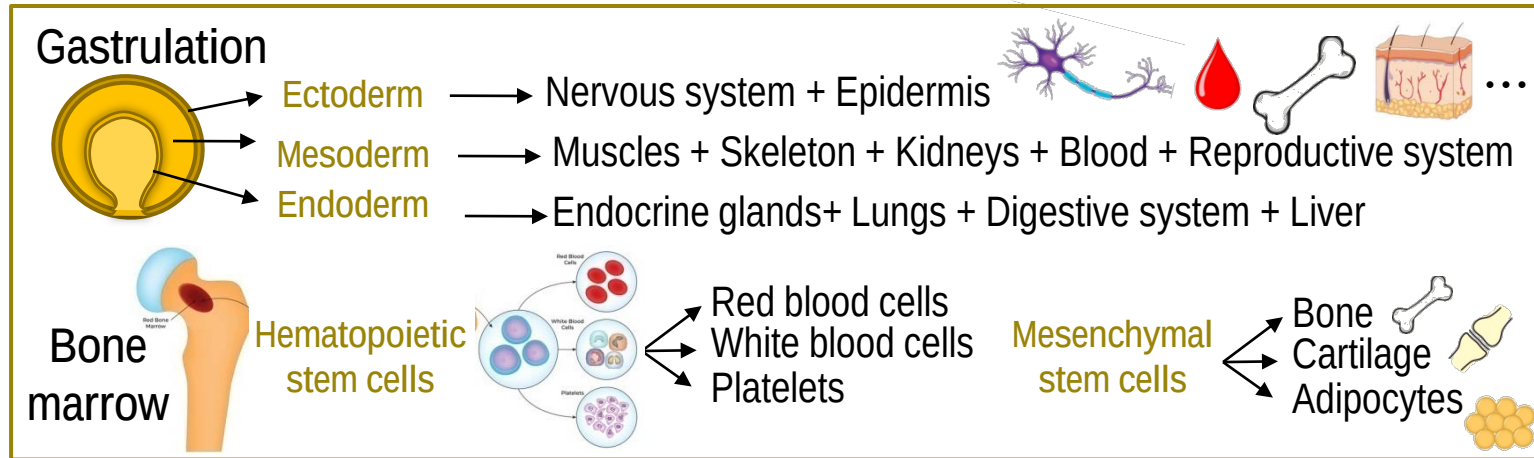
Gastrulation



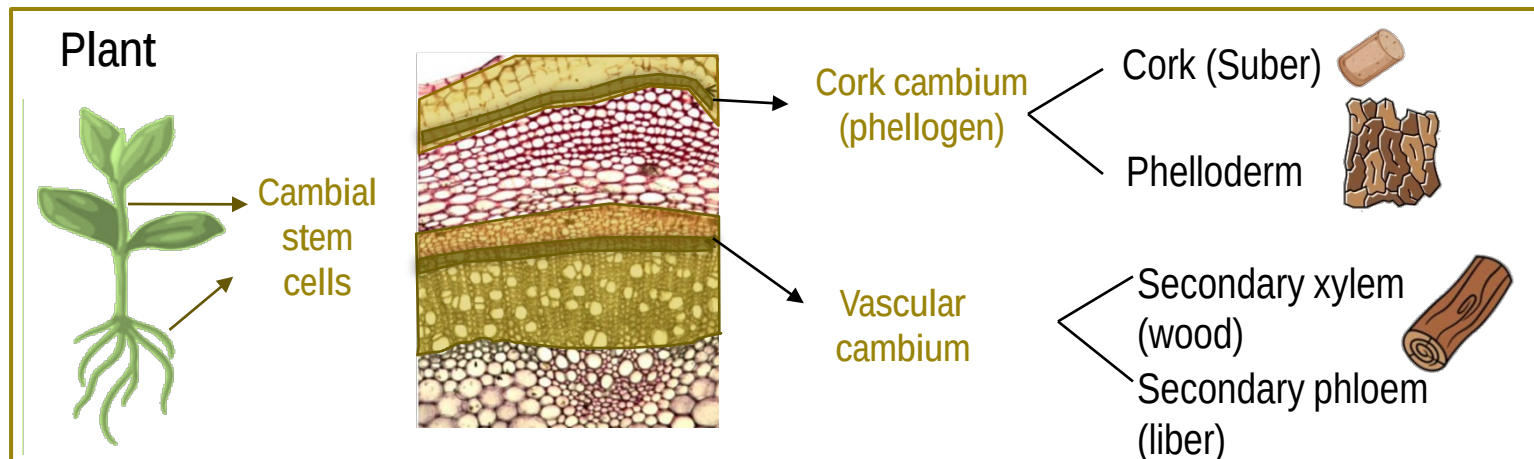
Multipotency



Totipotency of plants and animal cells



Multipotency



I

II

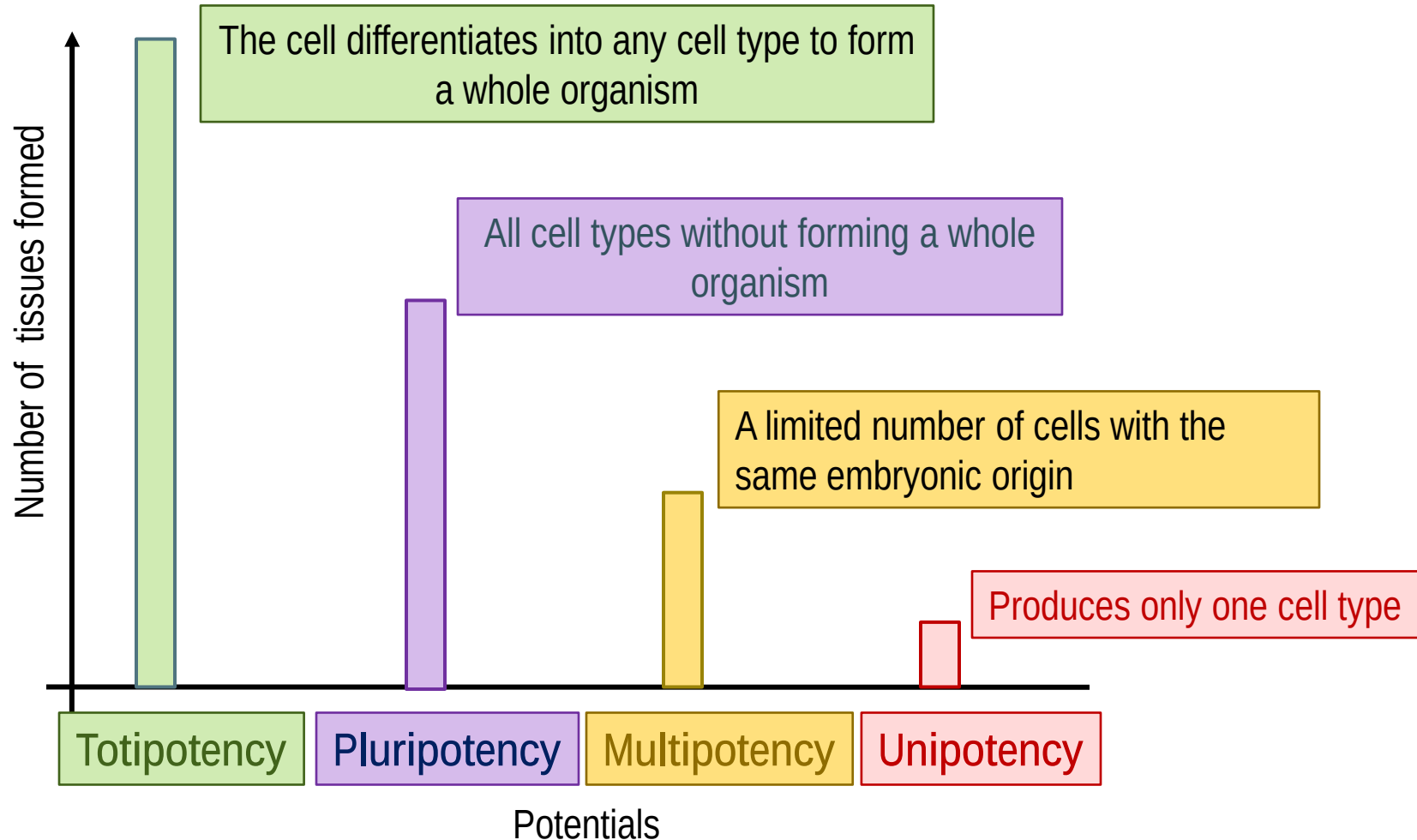
III

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VI

Totipotency and other cellular potentials



I

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Totipotency of plants and animal cells

Fertilization

Divisions

Morula

Blastocyst

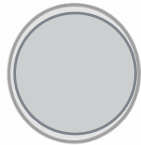
Gastrulation



Zygote



Early blastomeres
(2 and 4 cells)



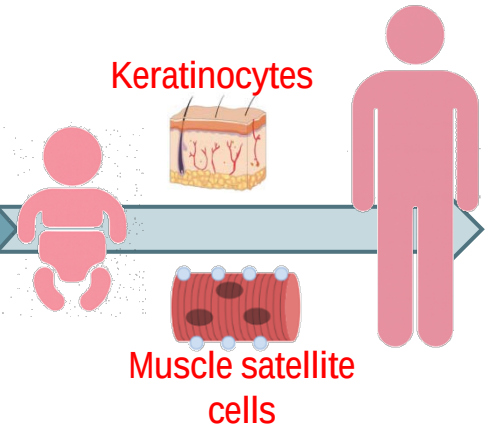
Embryo
(16 to 32 cells)



Inner cell
mass



Three embryonic
layers



Keratinocytes

Muscle satellite
cells

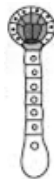
Unipotency



Zygote



Proembryo
(2 cells)



Globular
stage



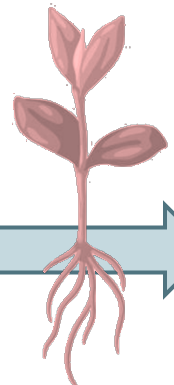
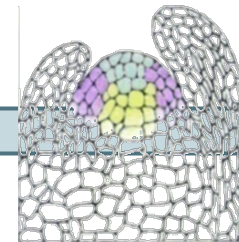
Heart
stage



Torpedo
stage



Pith meristem cells



I

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Totipotency of plants and animal cells

Keratinocytes

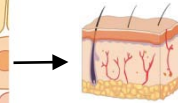
Stem cells
in the
basal layer

Stratum corneus

Stratum granulosum

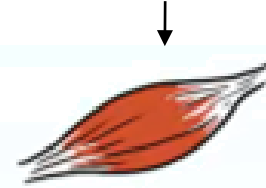
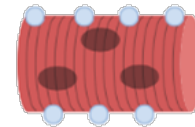
Stratum spinosum

Stratum basale



Epidermis

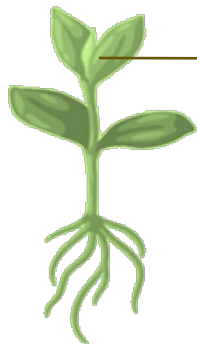
Muscle
satellite cells



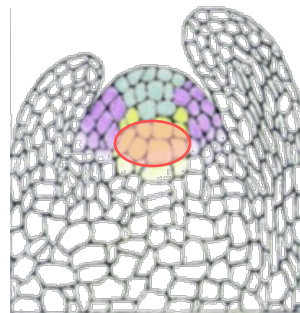
Skeletal muscle fibers

Unipotency

Pith



Meristematic
stem cells



Pith meristem

Stem pith



Sugarcane

I

II

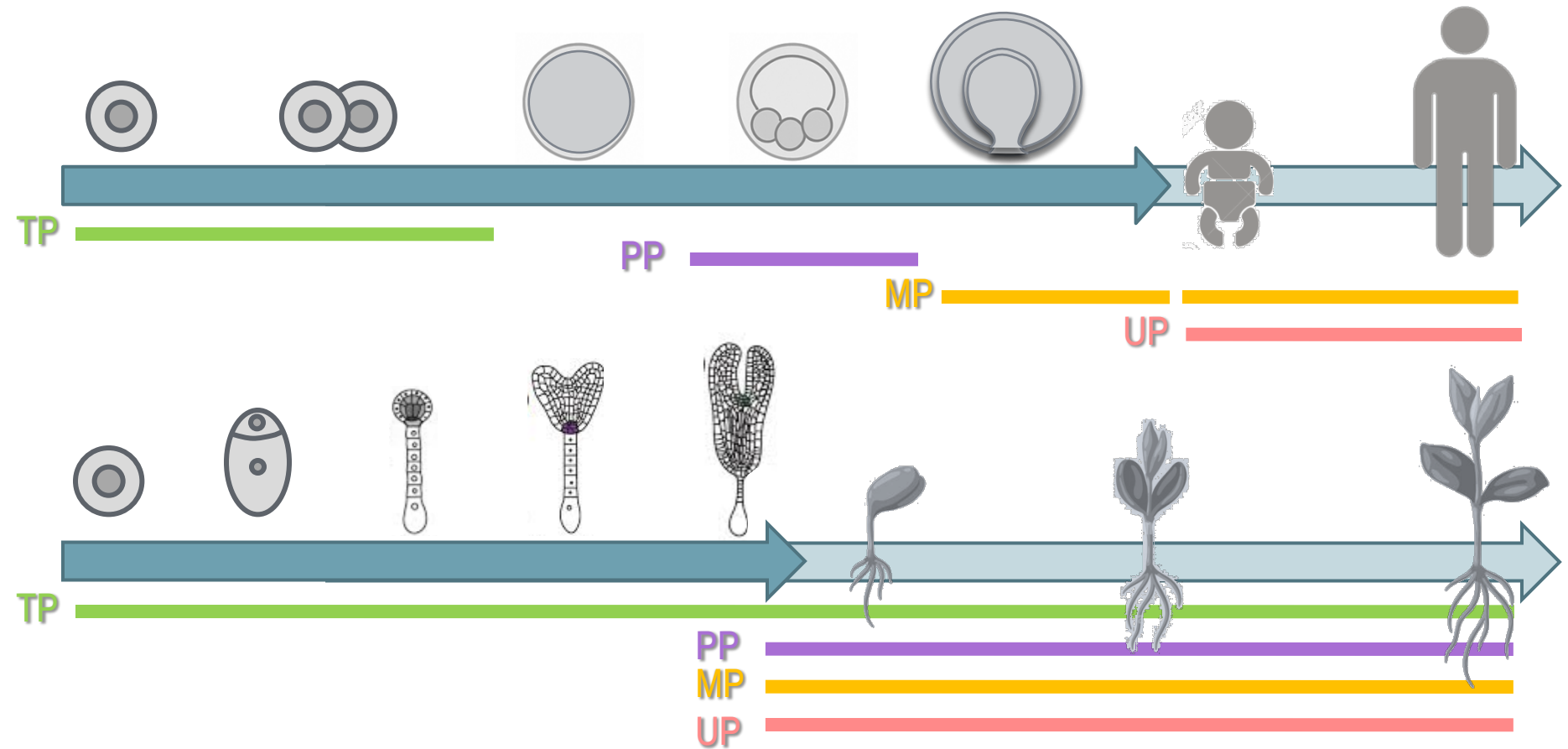
III

IV

V

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Summary



- Important post-embryonic development
- Formation of new organs throughout life → Organs of different ages
- Plasticity → Response to the environment
- High potential for cell culture

I

II

III

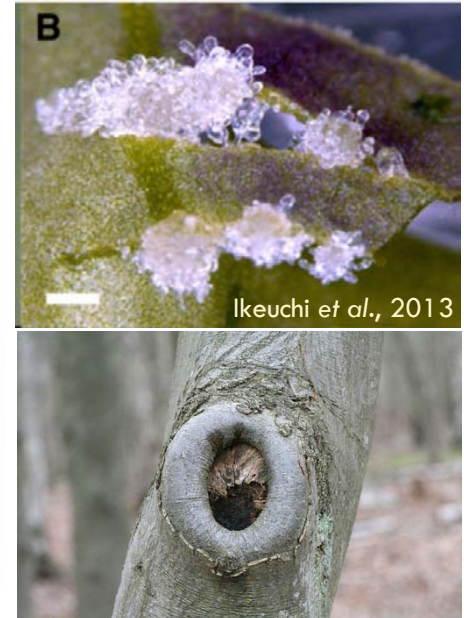
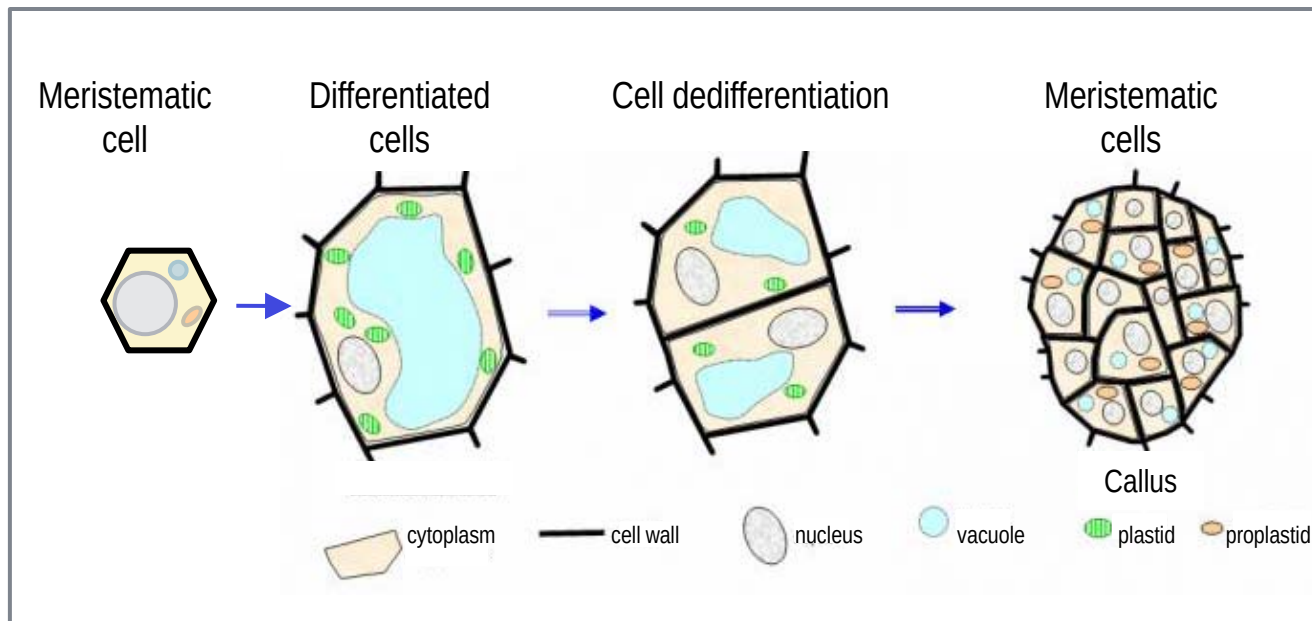
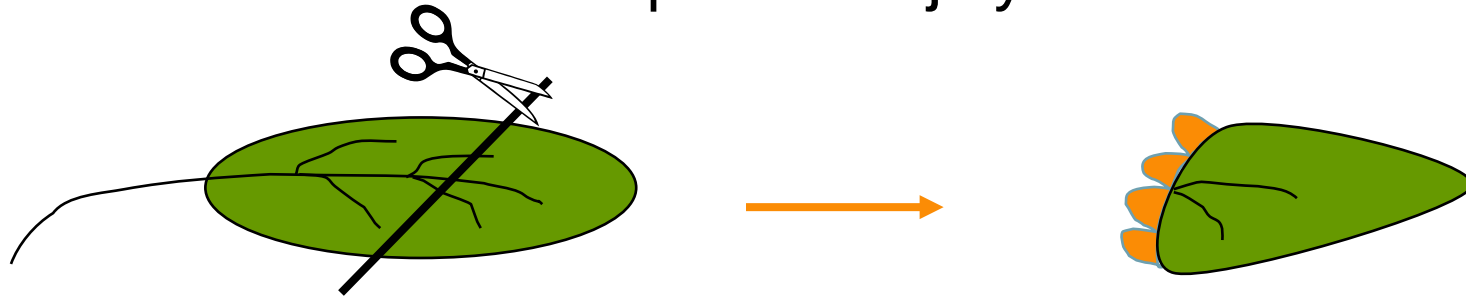
IV

V

VI

From cell dedifferentiation to callus

Response to injury

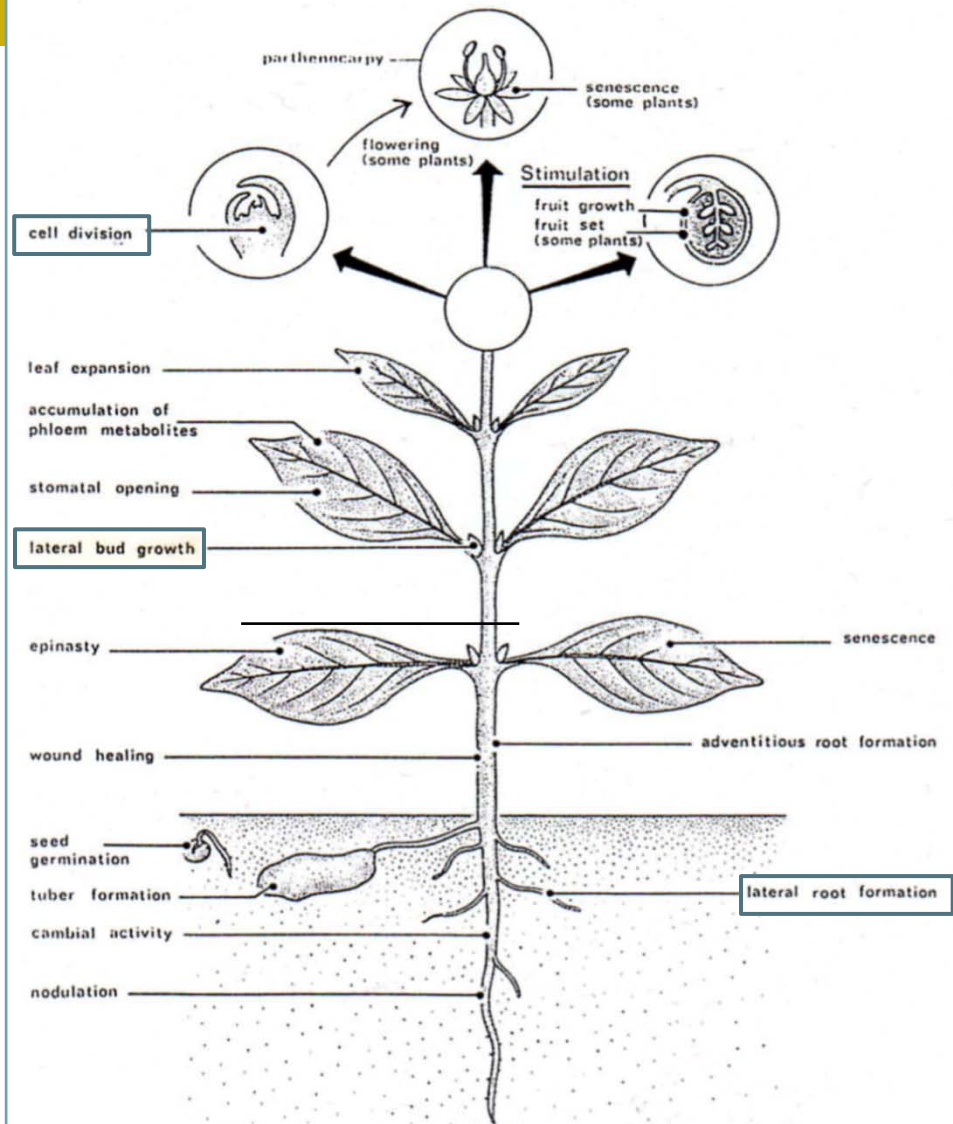


Pleiotropic hormones

Stimulation

Cytokinins

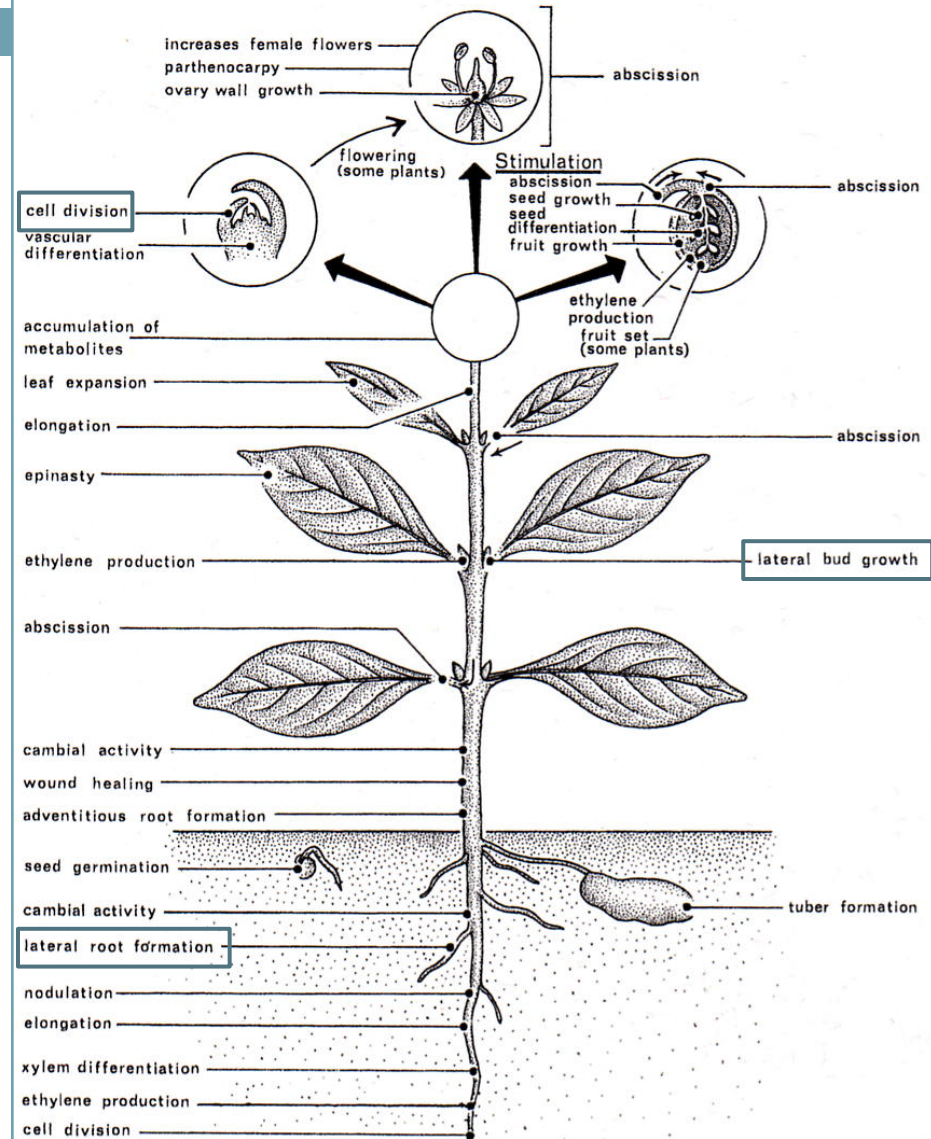
Inhibition



Stimulation

Auxins

Inhibition



I

II

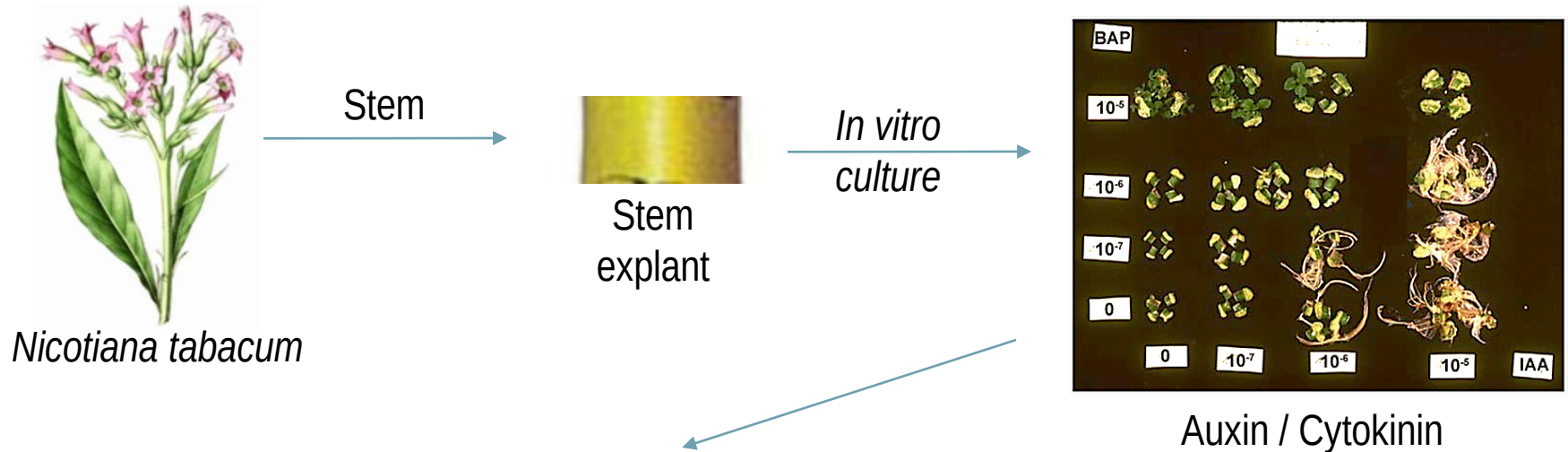
III

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Miller and Skoog experiment (1957)



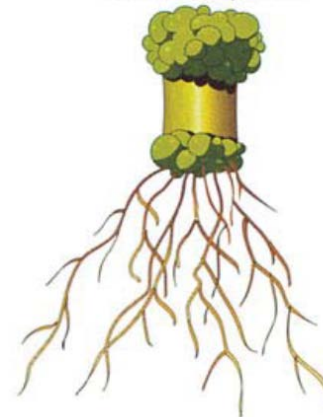
Caulogenesis



Callogenesis



Rhizogenesis



I

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Miller and Skoog experiment (1957)

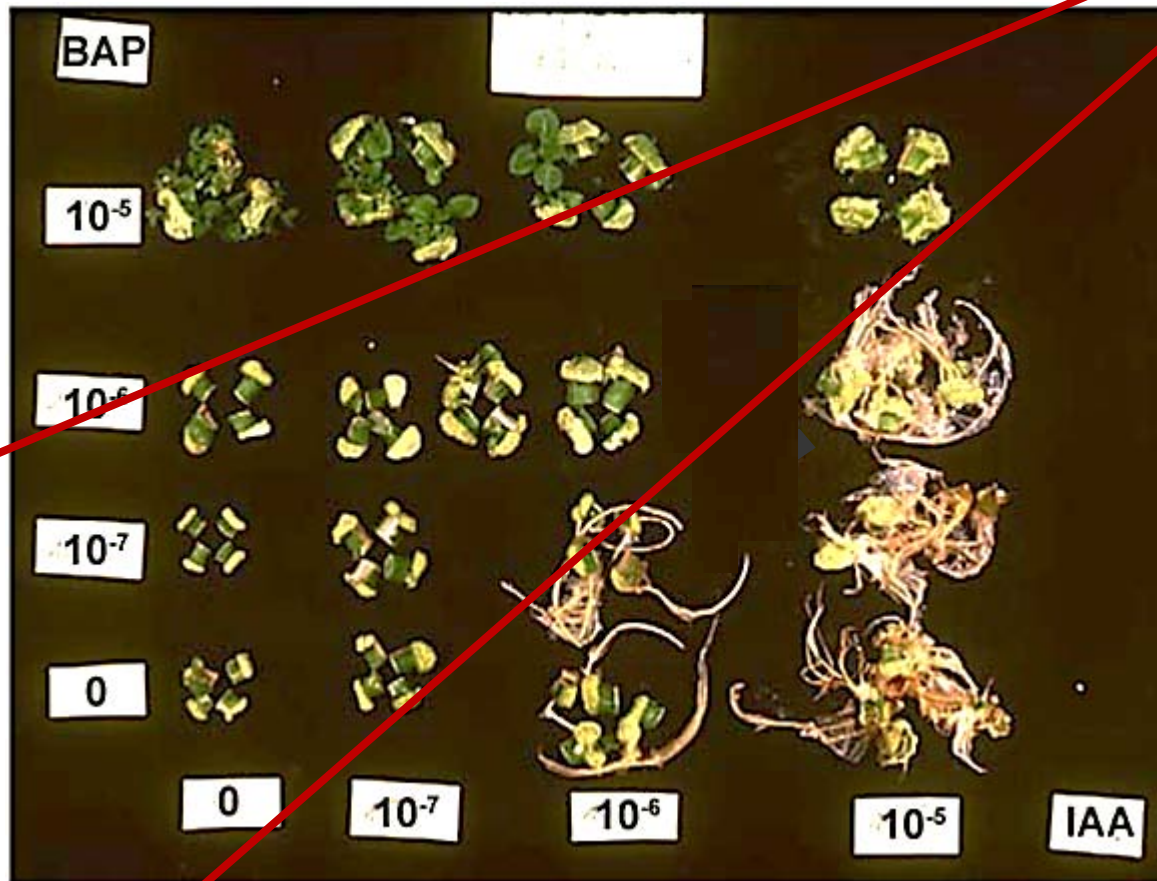
Cytokinin
(mol/L)

Caulogenesis

Aux / Ck $\searrow$

Callogenesis

Aux / Ck ~ 1



Rhizogenesis

Aux / Ck $\nearrow$

Auxin
(mol/L)

I

II

III

IV

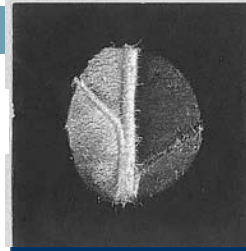
V

VI

Callogenesis

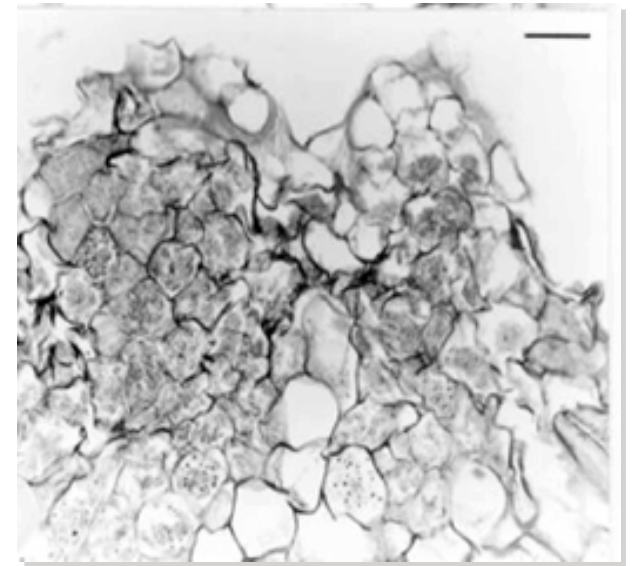
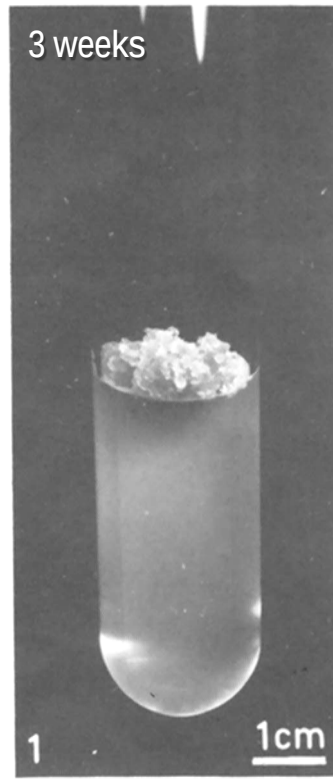


Datura nigra



Leaf explant

MS10 medium
+
Auxin 1 mg/l
Cytokinin 1 mg/l



I

II

III

IV

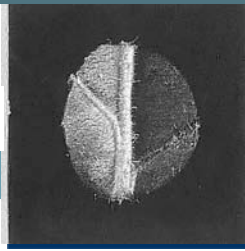
V

VI

Caulogenesis

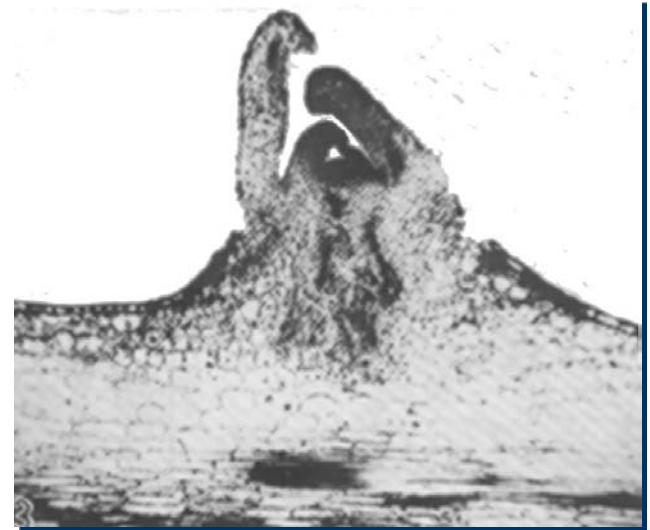
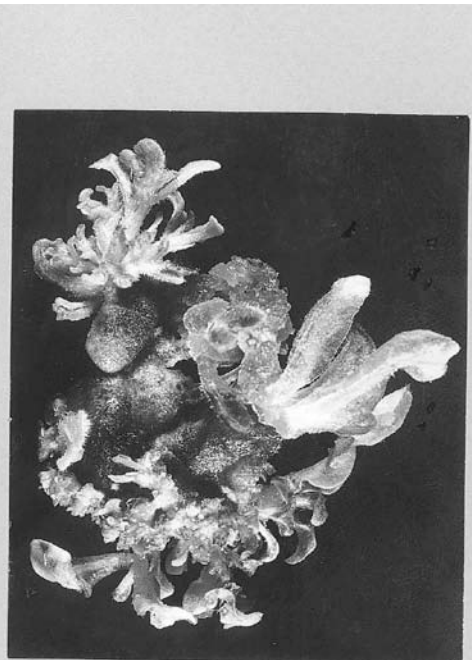
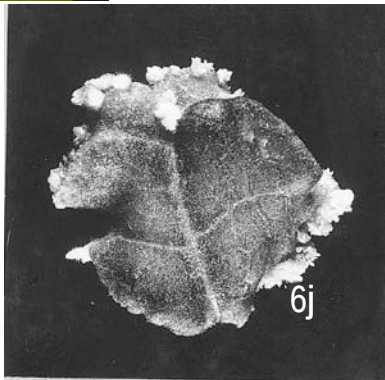


Nicotiana tabacum



Leaf explant

MS + AIA (0,2mg/l) + KIN (1mg/l)



I

II

III

IV

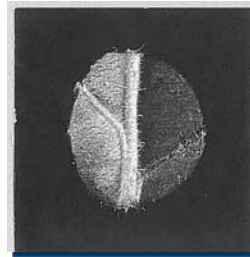
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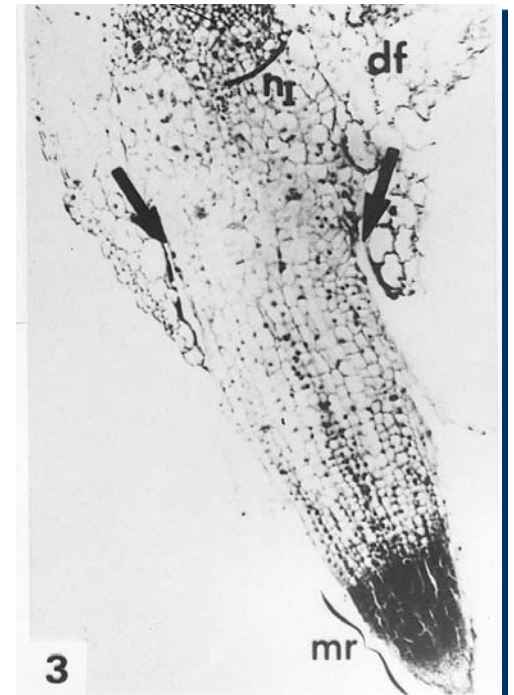
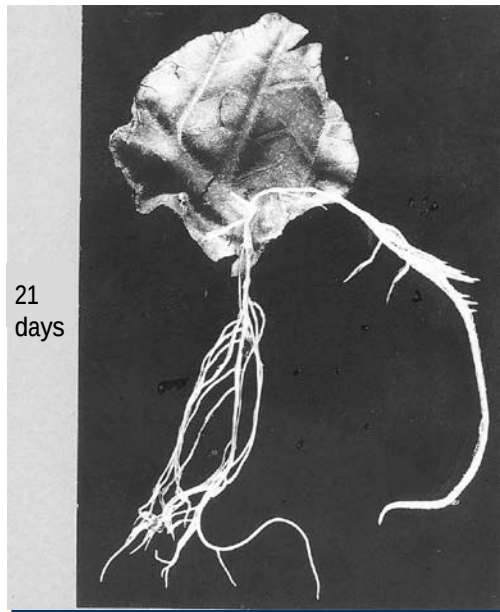
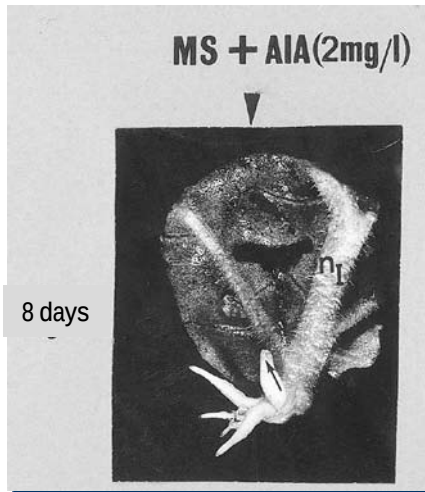
Rhizogenesis



Nicotiana tabacum



Leaf explant



I

II

III

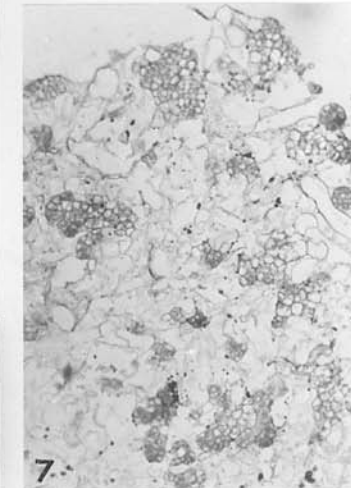
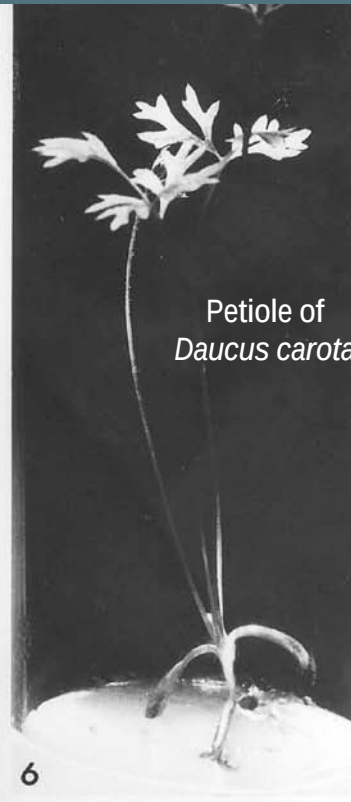
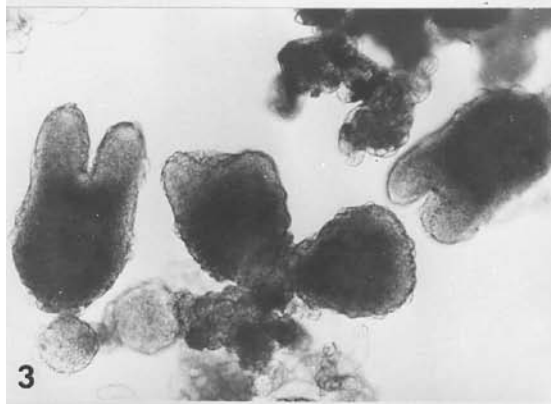
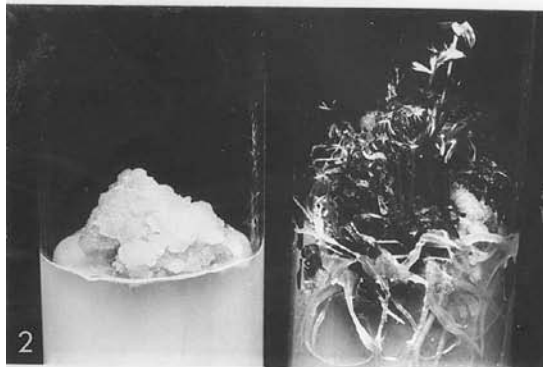
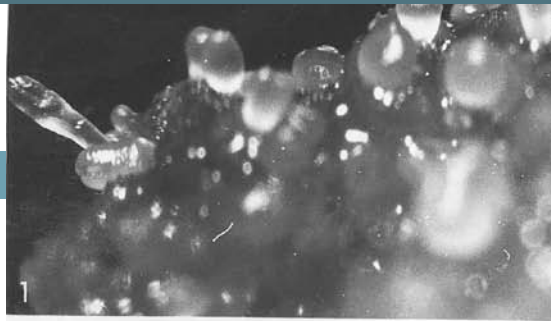
IV

V

VI

Somatic

embryogenesis



NITSCH + 2,4-D (1 mg/l)

I

II

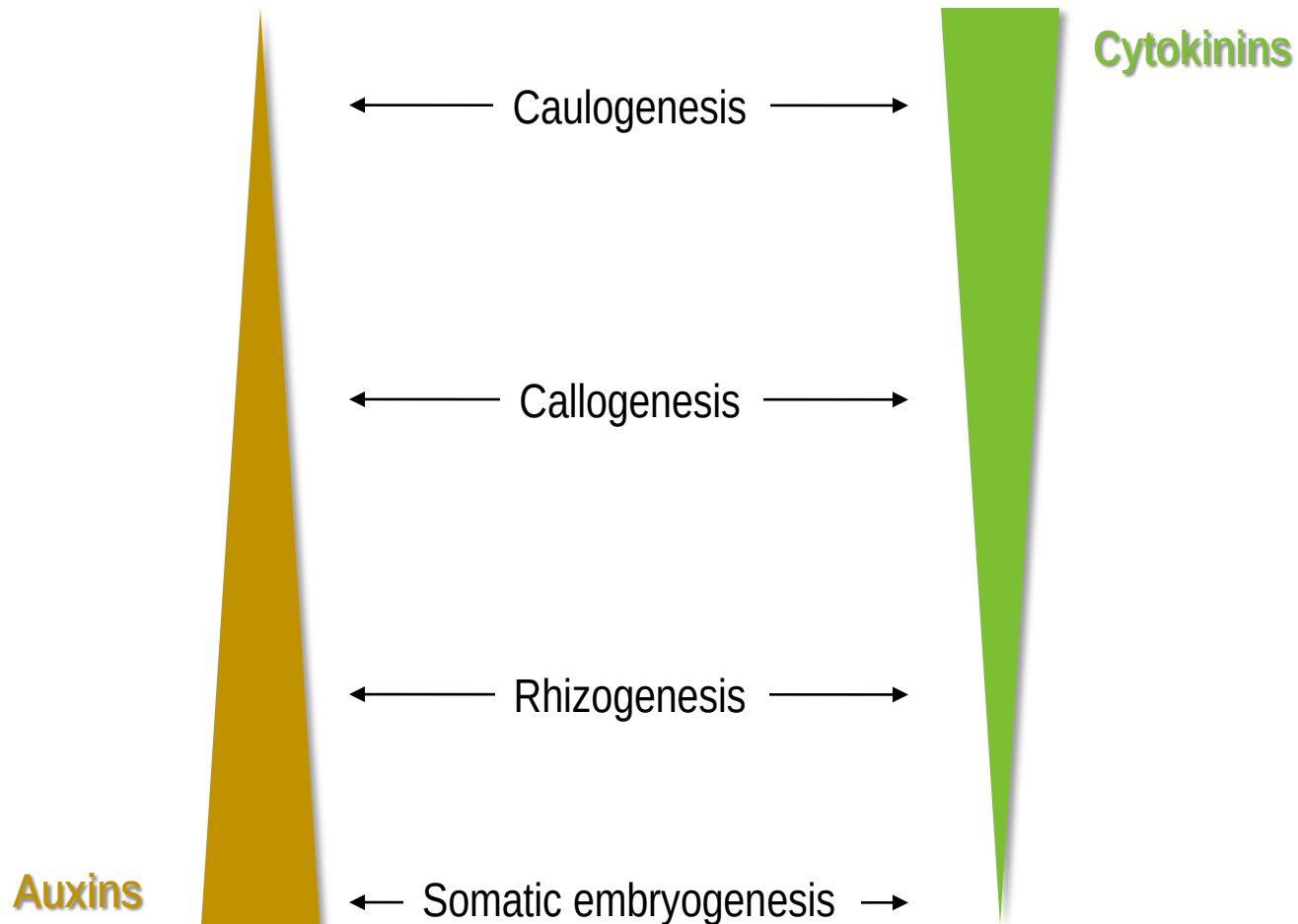
III

IV

V

VI

Role of hormones in reprogramming



I

II

III

IV

V

VI

Cellular
potentials

In vitro culture

Meristematic cell
(pluripotency)

Differentiation

Differentiated cell

Dedifferentiation

Dedifferentiated cell
(pluripotency, totipotency)

Reprogramming

Aux / Ck ~ 1

Aux (2,4-D) ↗ ↗

Callogenesis

Direct organogenesis

Direct somatic
embryogenesis

Aux / Ck ↘

Aux / Ck ↗

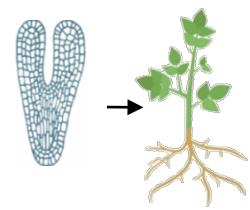
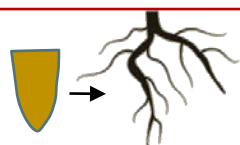
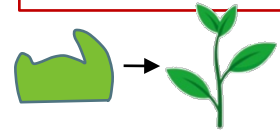
Caulogenesis

Rhizogenesis

Unipotency

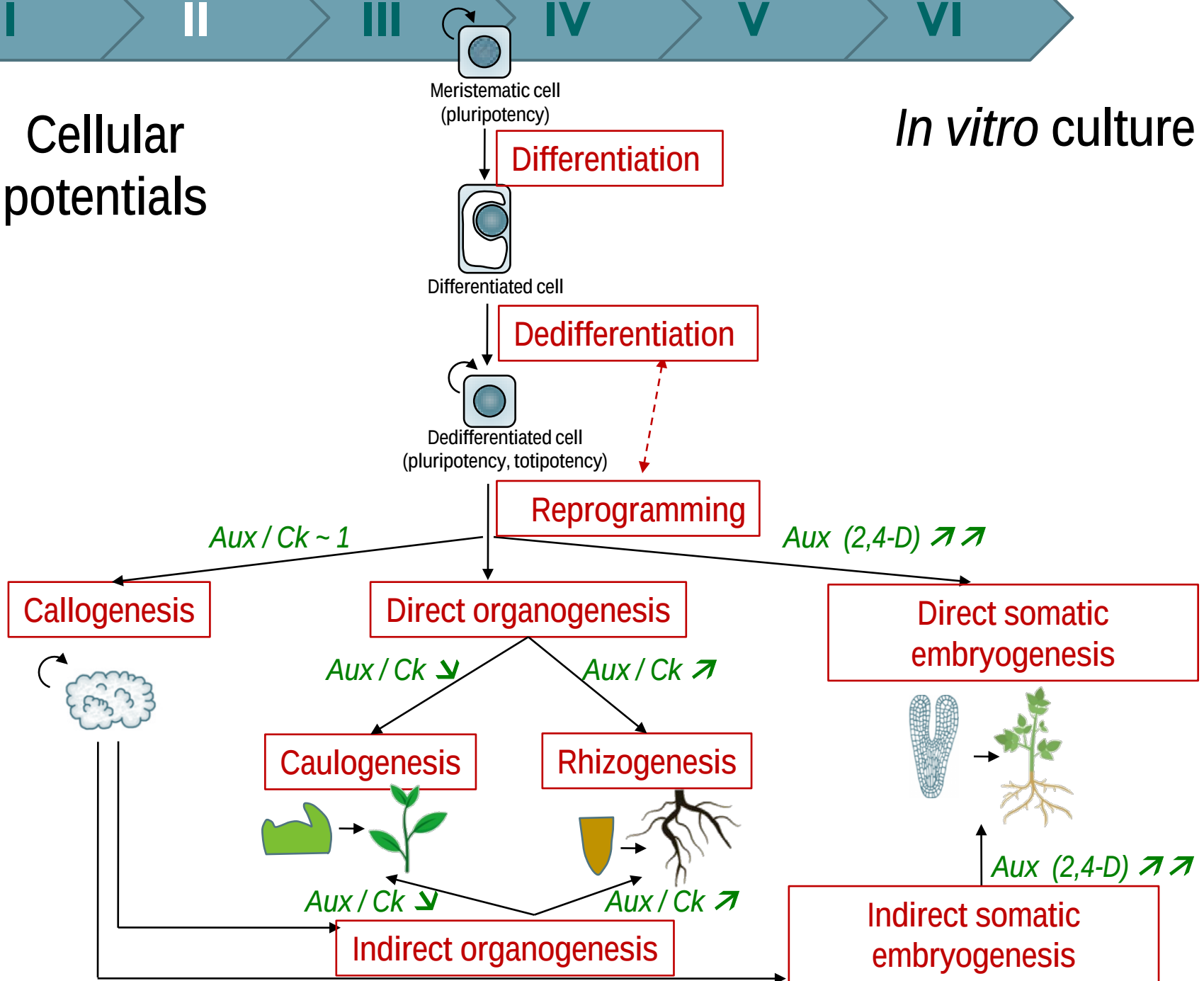
Pluripotency

Totipotency

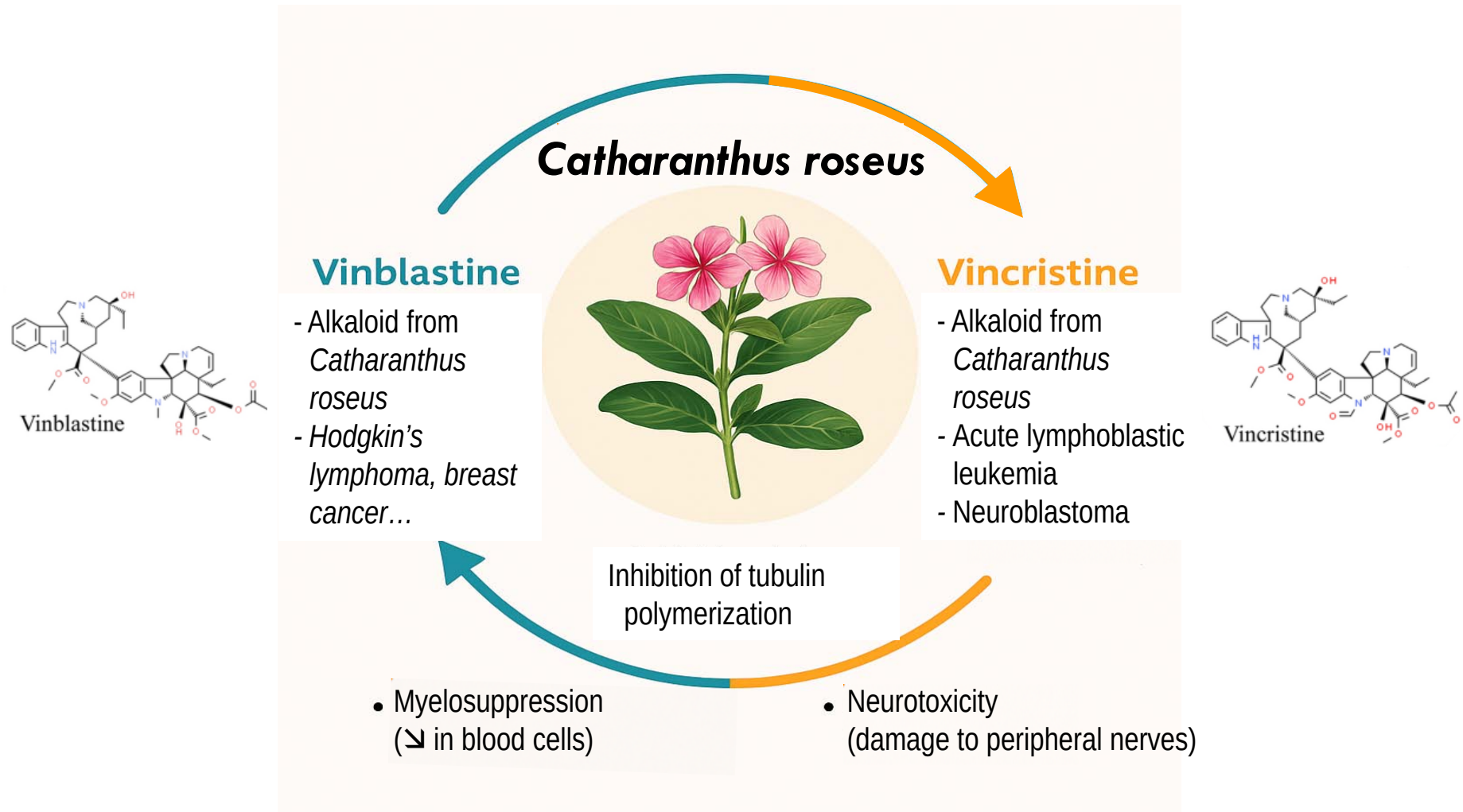


Cellular potentials

In vitro culture



Case study: the Madagascar periwinkle



Catharanthus roseus



Selection of the initial genotype



Initiation of cell lines



Primary callus

Stabilization of cell lines



Screening, selection

Cloning

Cell suspension



Optimization



Bioreactor culture

Development and definition of the process

Industrial production

Extraction

Vinblastine
Vincristine

Madagascar periwinkle (*Catharanthus roseus*)



- vinblastine 2-10 g per ton of fresh weight
- vincristine 0,5-1 g per ton of fresh weight



Elicitors : methyl jasmonates or sugars

Cell culture : x6-x20

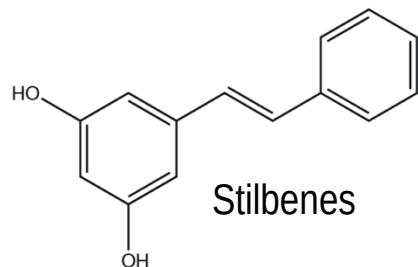


Case study: Fo-ti "Elixir of long life"

Traditional Chinese medicine



Fallopia multifolia,
(*Polygonum multiflorum*)

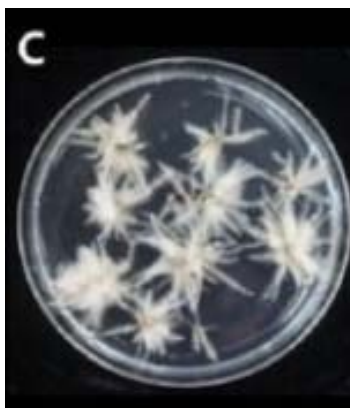


Neuroprotective properties

- Alzheimer
- Parkinson



Anthropomorphic root

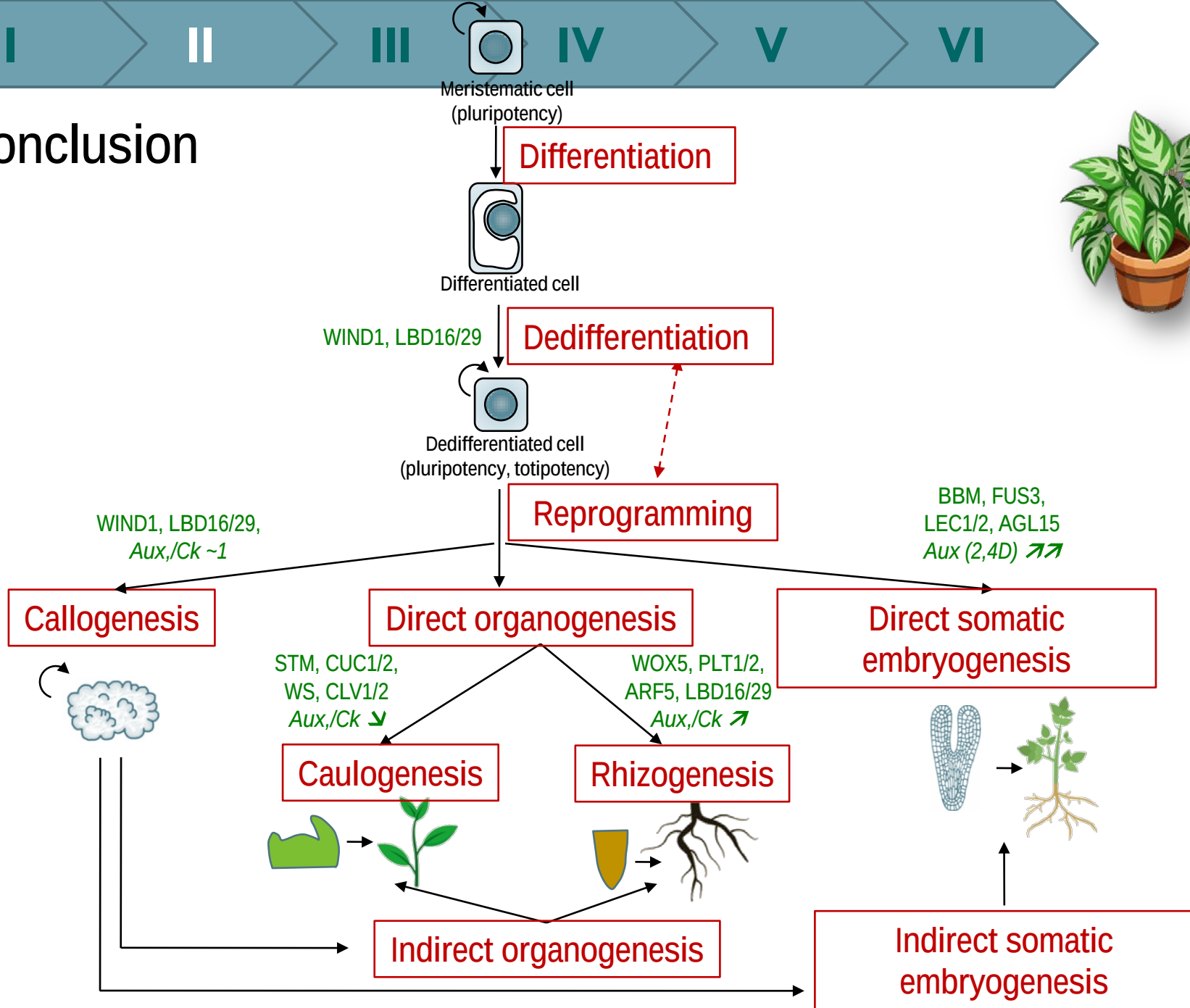


In vitro culture of adventitious roots



Bioreactor

Conclusion



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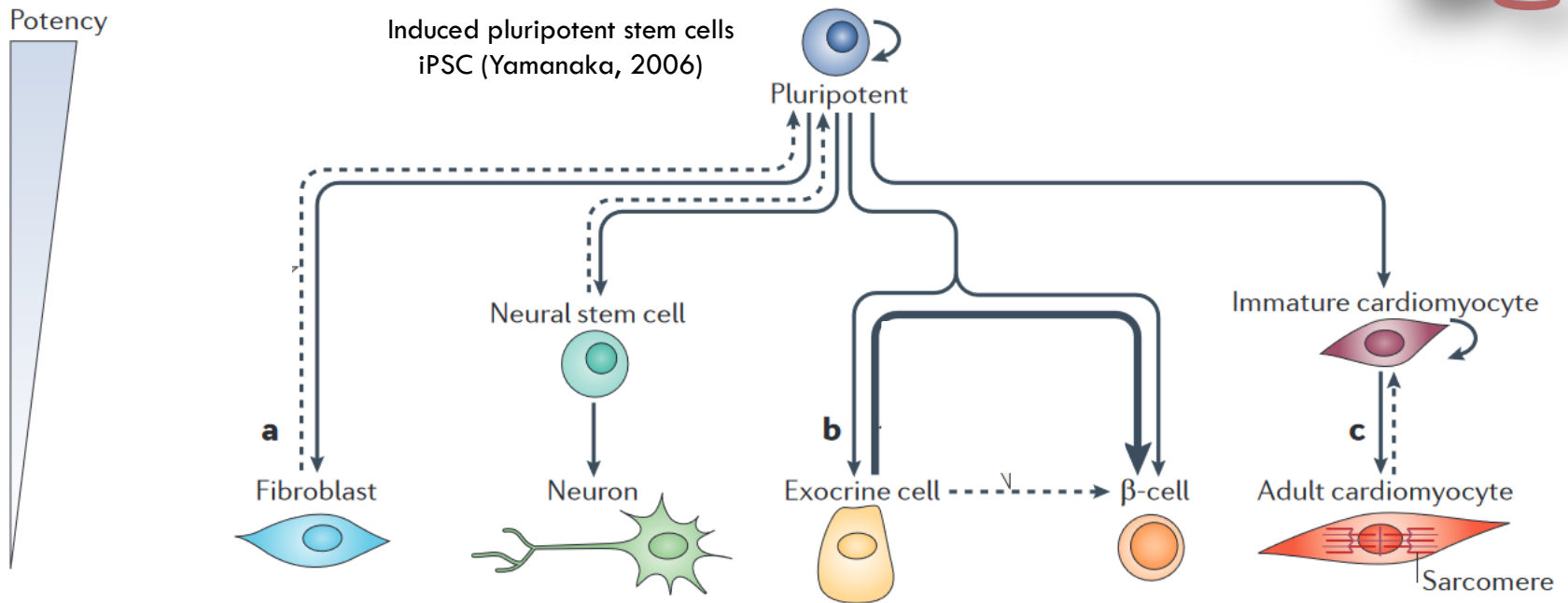
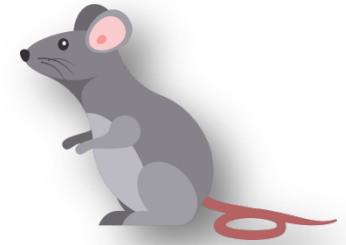
III

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Comparison



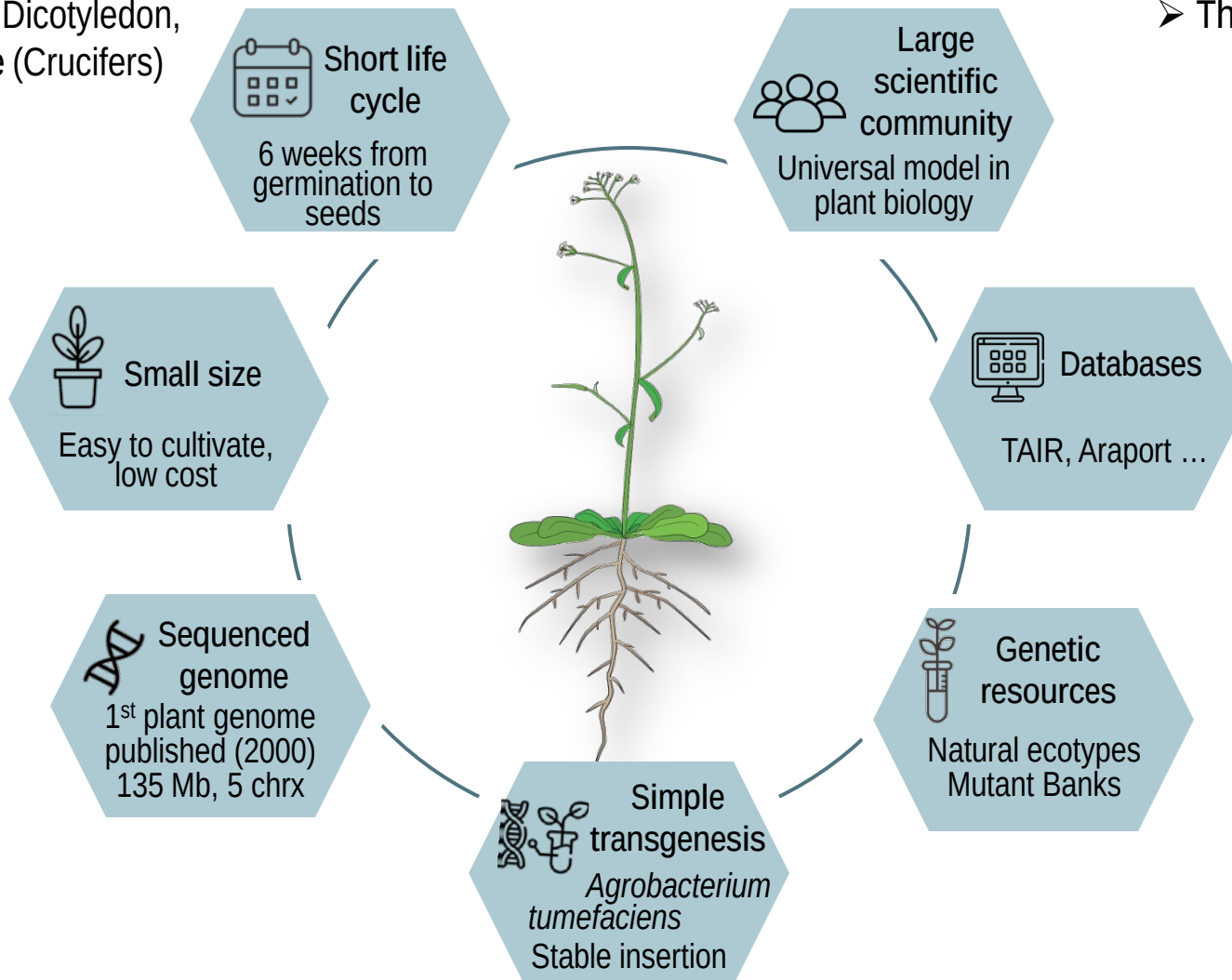
Course Outline

- I. Introduction
- II. Totipotency and plant cell models
- III. Model 1: *Arabidopsis thaliana* – “Physiological and developmental model”
- IV. Model 2: *Oryza sativa* – “Agricultural and genomic model”
- V. Model 3: *Populus trichocarpa* – “Woody plant model”
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Arabidopsis thaliana – Why this model?

➤ Angiosperm, Dicotyledon,
Brassicaceae (Crucifers)

➤ Thale cress (weed)



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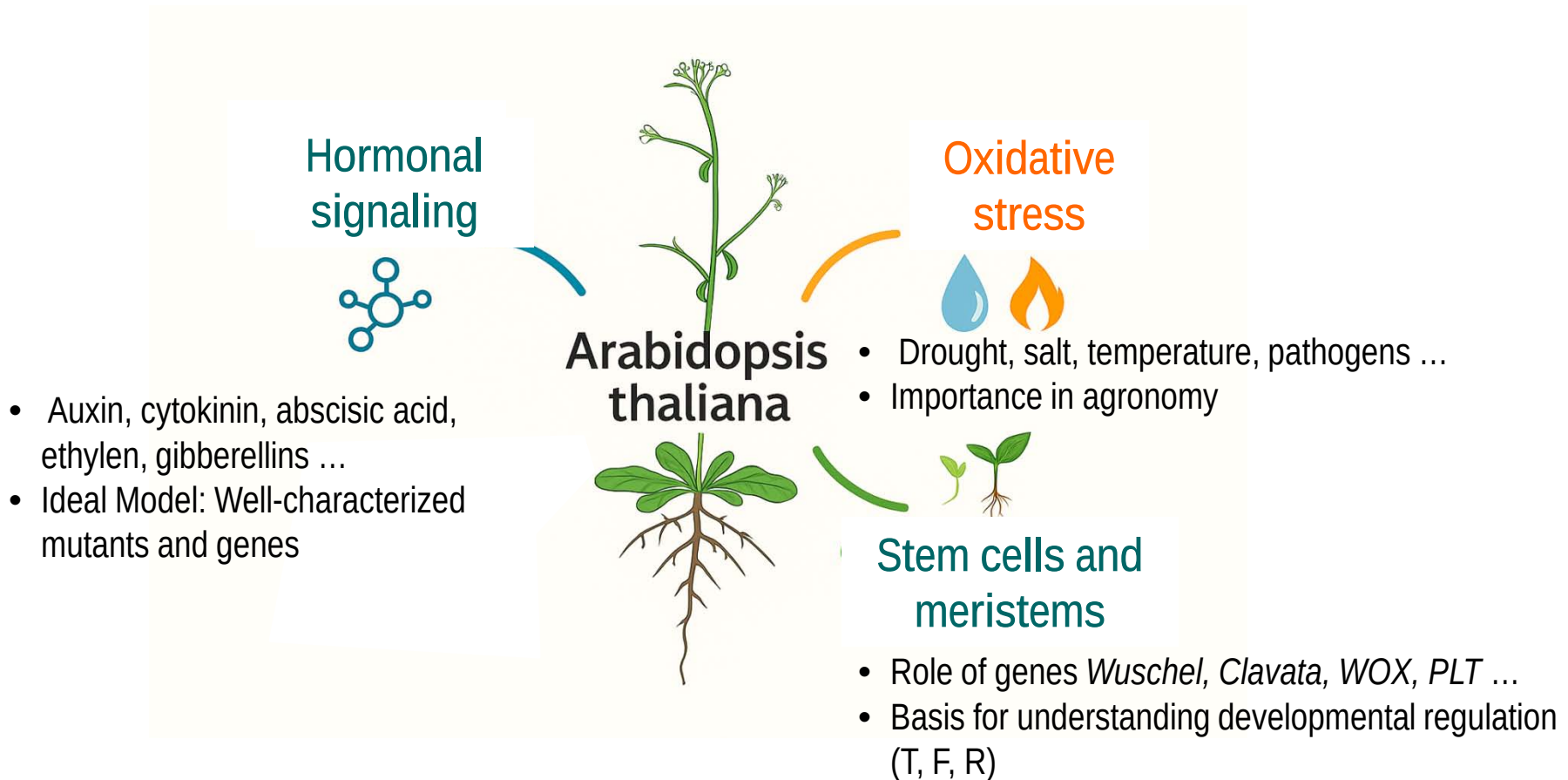
Arabidopsis thaliana – Why this model?

Plant	Genome size (millions bp)
Arabidopsis	135
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Poplar	500
Medicago (Alfalfa)	650
Cabbage	700
Sorghum	800
Rapeseed	1 200
Maize	2 500
Tabacco	5 000
Pea	5 200
Wheat	16 000

3 200 Mb



Arabidopsis thaliana – Main applications



I

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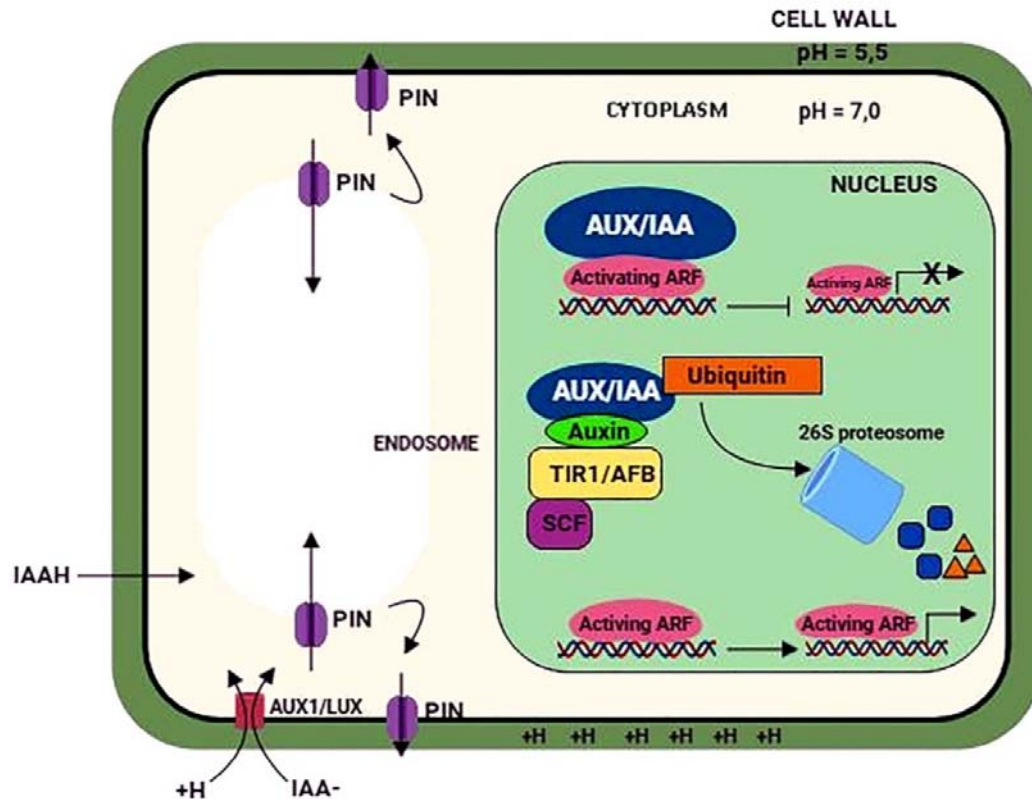
IV

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VI

Case study - Hormonal signaling

Auxin

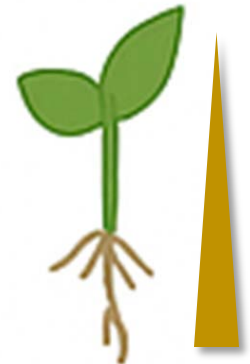


Target gene
expression

Shoot
elongation ...

Cell division

Rhizogenesis



Pantoja-Guerra et al., 2023

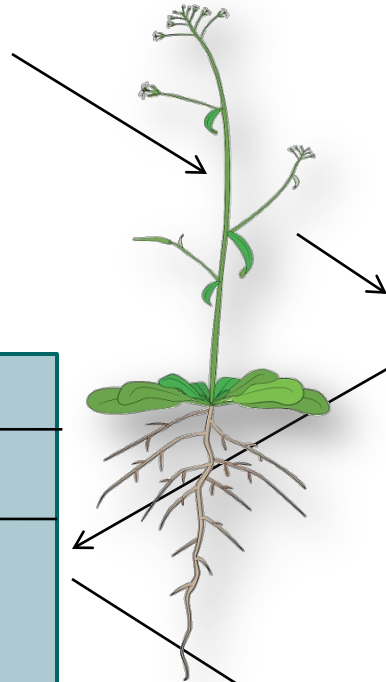
Study of root and shoot developmental control

Case study - oxidative stress

Stress



Drought, salt,
temperature,
pathogens



Reactive oxygen species (ROS)

- Superoxide anion O_2^-
- Hydrogen peroxide H_2O_2
- Hydroxyl radical $\cdot O^-$

Antioxidant defenses

Enzymatic reactions

- SOD (Superoxide dismutase)
- CAT (Catalase)
- APX (Ascorbate peroxidase)
- GPX (Glutathione peroxidase)
- GR (Glutathione reductase)

Non-enzymatic reactions

- Ascorbate (Vitamin C)
- Glutathione (GSH)
- Carotenoids (b-carotene, lutein)
- Tocopherol (Vitamin E)
- Flavonoids and polyphenols

Responses

Signaling

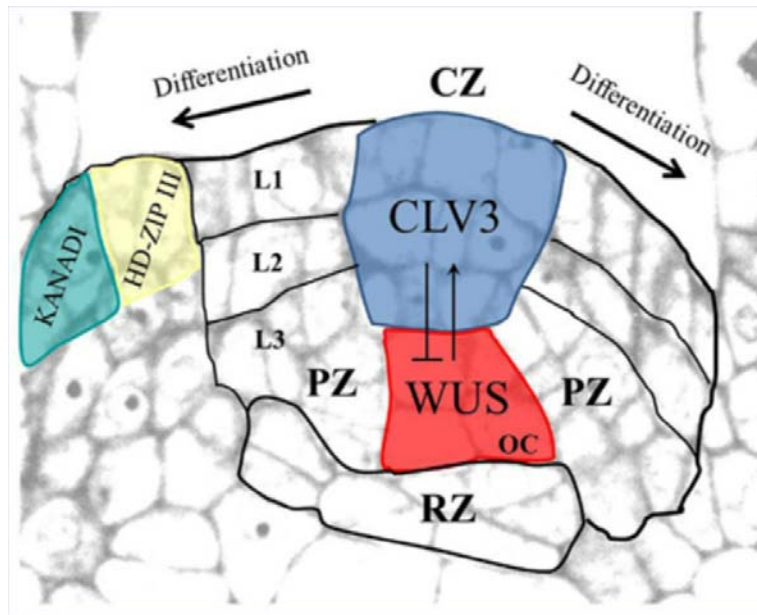
- Activation defense pathways (stress response)
- Growth regulation (roots...)
- Cellular and hormonal communication (auxin...)
- Physiological adaptation (tolerance to drought, salt ...)

Damages

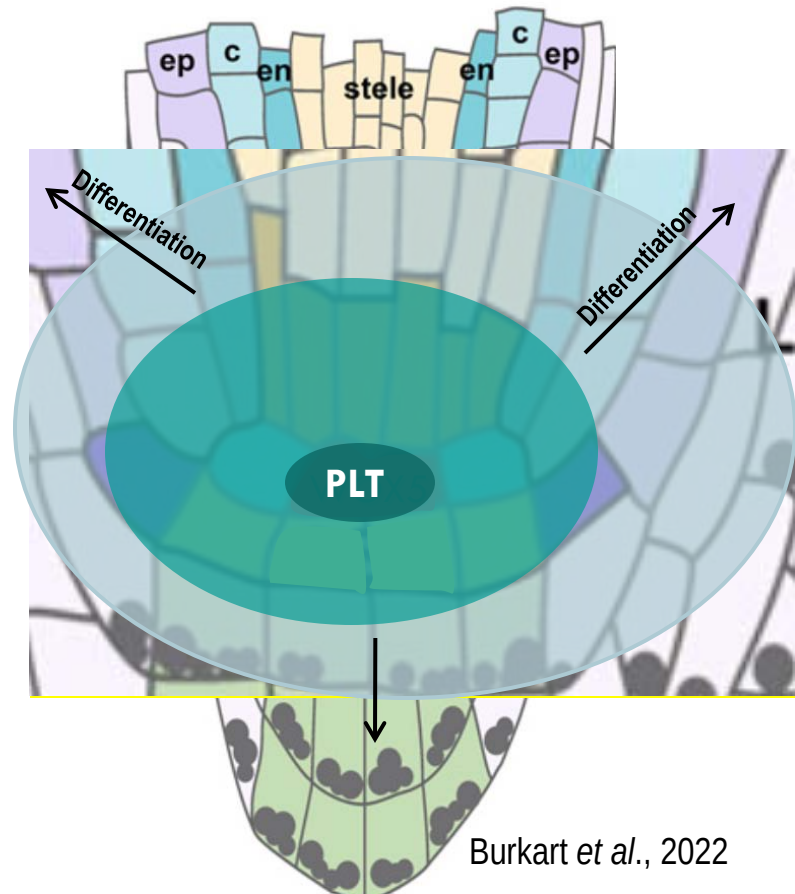
- Membrane lipid peroxidation → **loss of cell integrity**
- Protein oxidation → **loss of enzymatic activity**
- DNA mutations/breaks
- Cell death

Arabidopsis has therefore revealed how ROS transform stress into an adaptive signal

Case study - Stem cells



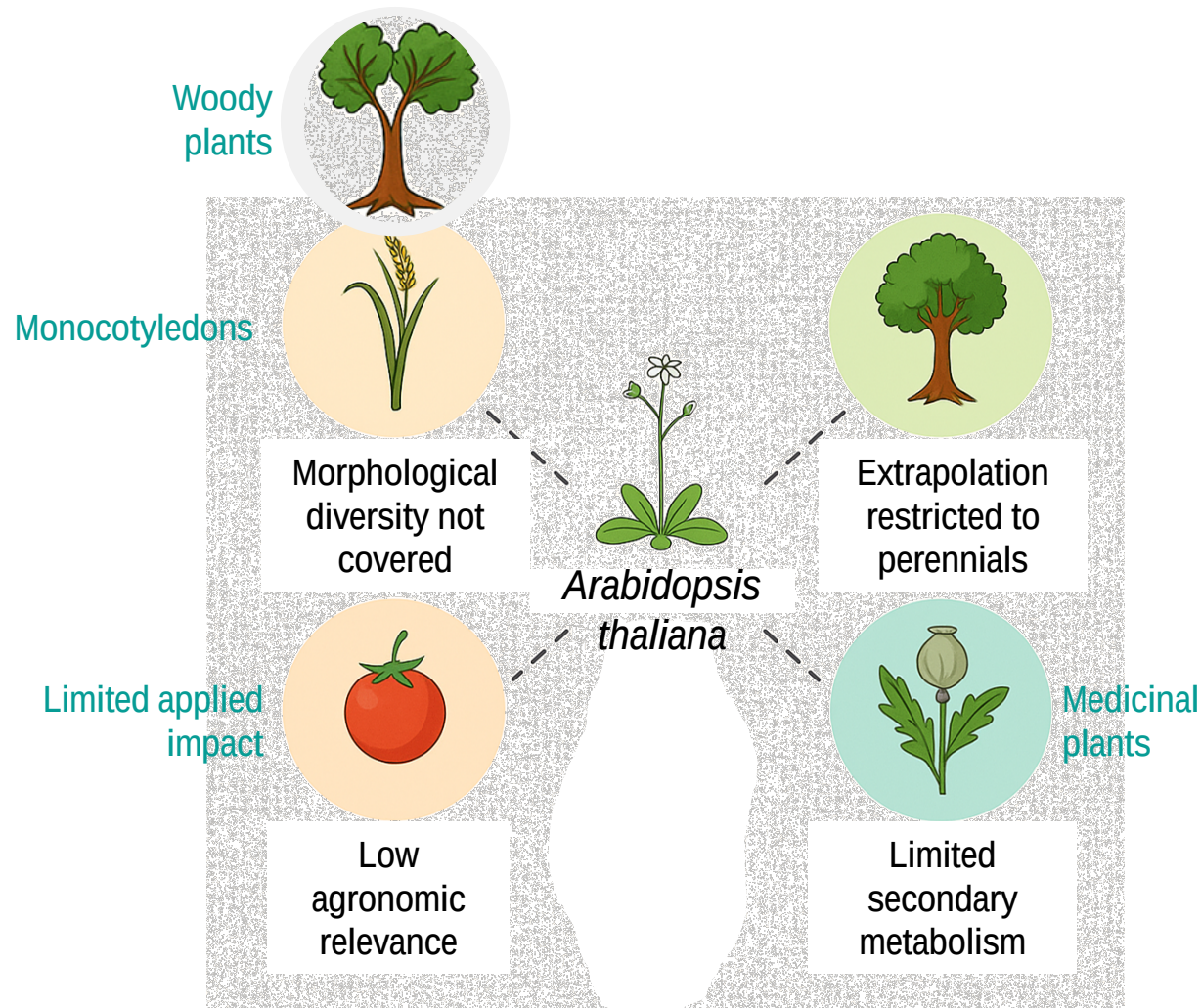
<https://bcgdevelop.fr/>



Burkart et al., 2022

Arabidopsis is a good model for understanding how stem cells regulate growth and organ formation

Arabidopsis thaliana - Limitations of the model

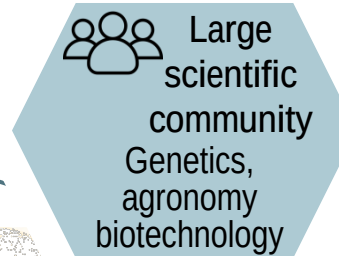
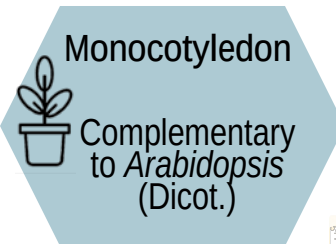


Course Outline

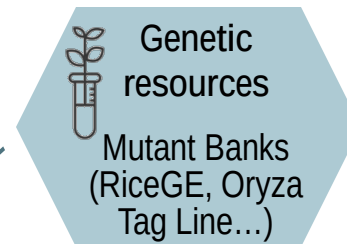
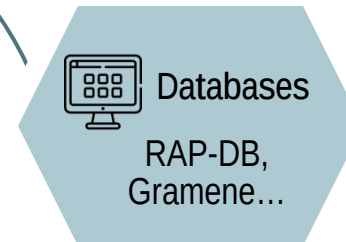
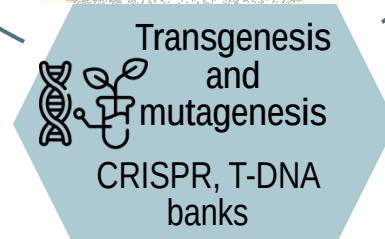
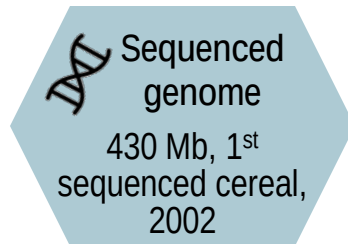
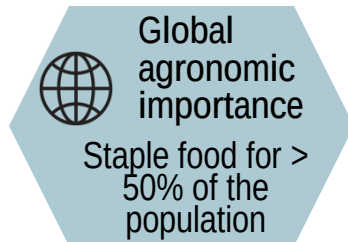
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Oryza sativa - Why this model?

➤ Angiosperm,
Monocotyledon, Poaceae
(Grasses)



➤ Most widely cultivated
and oldest cultivated plant
(20 000 rice varieties)
➤ High water requirement



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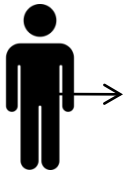
V

VI

Oryza sativa - Why this model?

Plant	Genome size (millions bp)
Arabidopsis	135
Rice	430
Poplar	500
Medicago (Alfalfa)	650
Cabbage	700
Sorghum	800
Rapeseed	1 200
Maize	2 500
Tabacco	5 000
Pea	5 200
Wheat	16 000

3 200 Mb



Oryza sativa – Main applications

Water stress



Upland rice (plateau rice, drought-resistant)

Conditions

- Dry soils, frequent water stress

Physiological adaptations

- Deep and branched roots -> better water absorption
- Fine regulation of transpiration -> better control of stomata

Key genes

- DREB (transcription factor activating drought-tolerance genes)
- NAC (metabolic adaptation during drought periods)
- Aquaporins (transporters facilitating water absorption)



Lowland rice (flood-tolerant rice)

Conditions

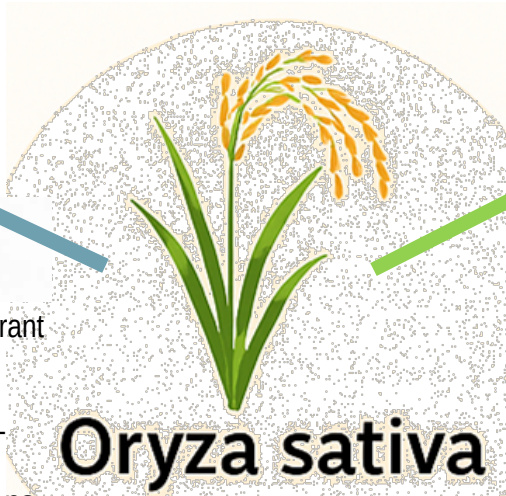
- Flooded paddies, oxygen-poor soils

Physiological adaptations

- Aerenchyma -> tissue enabling oxygen transport
- Hypoxia tolerance

Key genes

- SUB1A (hypoxia-response gene, important for survival under submergence)
- Aquaporins (maintaining water flux)



Agronomy

- Yield-related traits (grain number, panicle size)
- Nutritional quality (proteins, starch, micronutrients)

Biotechnology

- Salt-tolerant rice
- Introduction of *Saltol* (salt-tolerance QTL / chromosome 1)
- Multiples genes (Exclusion of Na⁺ from sensitive tissues + K⁺/Na⁺ Selectivity + Hormonal regulation under stress)
- Improves rice cultivation in coastal areas (International Rice Research Institute, Philippines)

Case study – Golden Rize

Problem

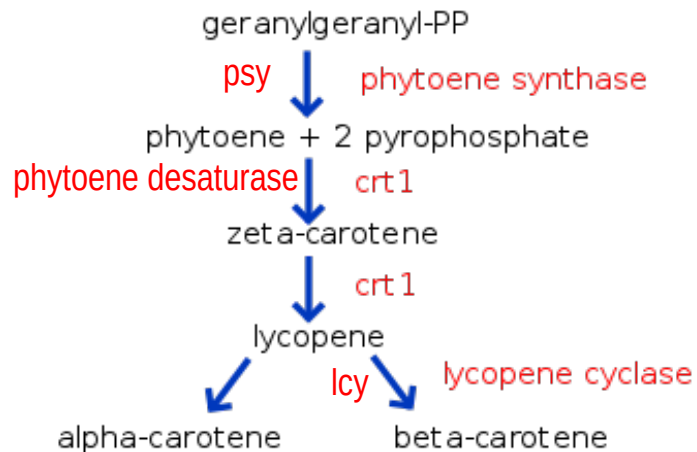


Vitamin A
deficiency



Beyer et al., 2002

Genetic approach



- *psy* - Narcisse
promo. GT1 endosperm-specific
- *crt1*- *Erwinia uredovora*
promo. 35s of CaMV
- *lcy*- Narcisse
promo. GT1 endosperm-specific
- Plastid targeting signal

Impact



- $\approx 1,6 \mu\text{g}$ β -carotene / g dried rice
100 g of rice $\approx 160 \mu\text{g}$ β -carotene
- Daily requirements : $600 \mu\text{g}$



Nutritional
improvement

Transgenesis can transform a staple food into a public health tool

Oryza sativa - Limitations of the model

Arabidopsis vs Rice: experimental differences



Arabidopsis

Laboratory model

- Short life cycle (= 6 weeks)
- Small size, easy to cultivate
- Homogeneous genome, low diversity
- Low logistical requirements (growth chambers)



Riz

Agricultural model

- Long life cycle (= 4 – 5 months)
- Demanding cultivation (water, space, greenhouse or field)
- Varietal diversity (*indica* vs *japonica*)
- Heavy logistical constraints (farming, experimental fields)

Arabidopsis → simplicity —————→ Rice → complexity

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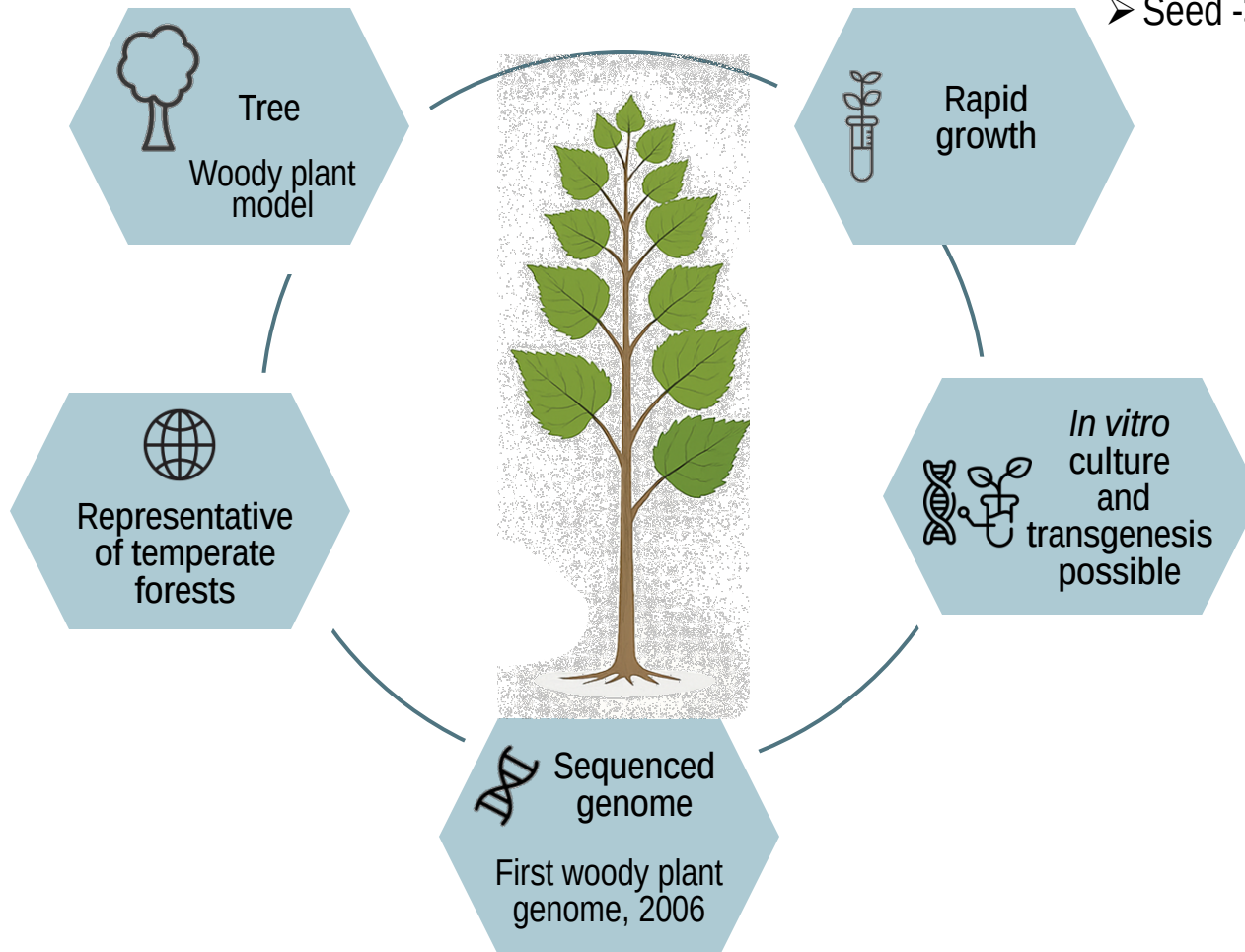
V

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Populus trichocarpa – Why this model ?

➤ Angiosperm, Cotyledon,
Salicaceae

➤ Woody plant model
➤ Fast-growing / 60 years
➤ Seed -> flowering ~7 years



Model 3 – Poplar – *Populus trichocarpa*

Plant	Genome size (millions bp)
Arabidopsis	135
Rice	430
Poplar	500
Medicago (Alfalfa)	650
Cabbage	700
Sorghum	800
Rapeseed	1 200
Maize	2 500
Tabacco	5 000
Pea	5 200
Wheat	16 000



Populus trichocarpa - Main applications



Applications of *Populus*

Wood
& lignification



Biomass
& bioenergy



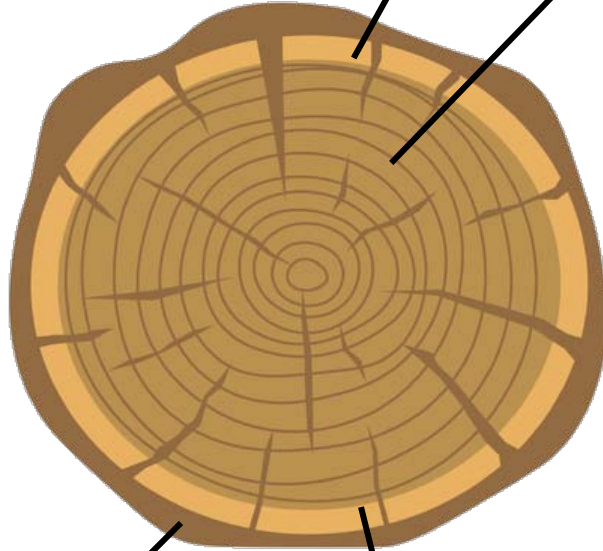
Chronic
stress



Case study - Lignification

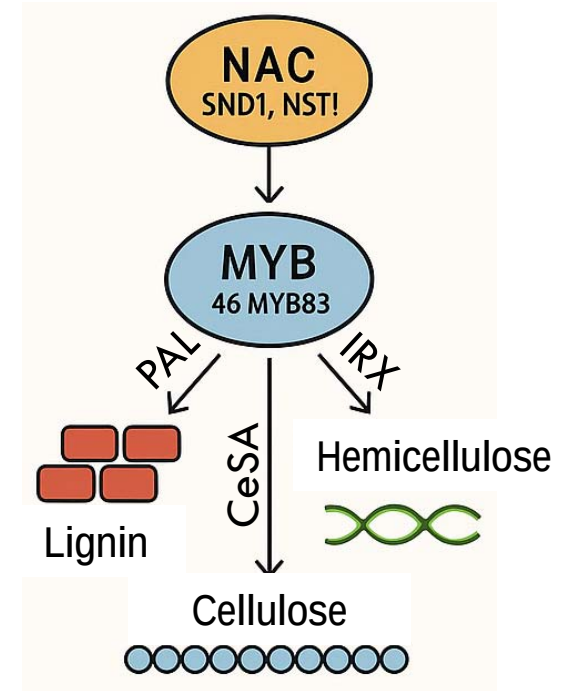
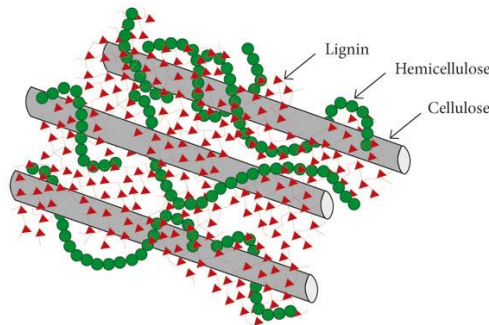
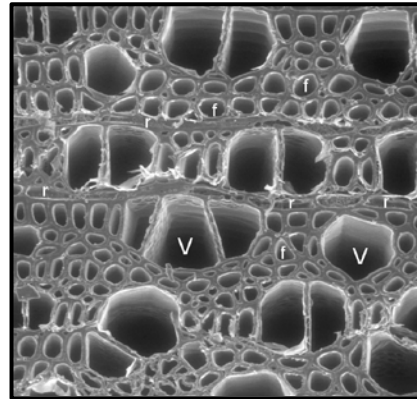
Secondary phloem (bast)
Elaborated sap
(sugars)

Secondary xylem (wood)
Raw sap
(water and mineral salts)



Bark

Cambium
(secondary
meristem)



Modulation of gene expression:

- ↗ wood rigidity: Industry
- ↘ lignin: bioenergy

Populus trichocarpa - Limitations of the model



Genetic study

Too long to wait for
F1, F2 ... generations
(flowering after 7 years)



In vitro culture

True-to-type micropropagation
→ genetic clones

Transgenesis

OGM, mutants

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Conclusion

	 <i>Aradidopsis thaliana</i>	 <i>Oryza sativa</i>	 <i>Populus trichocarpa</i>
Life cycle	Very short (6–8 weeks)	Intermediate (4 - 5 month)	Very long (7 years before flowering)
Plant type	Herbaceous dicotyledon	Monocotyledon (cultivated grass)	Woody dicotyledon (fast-growing)
Genome size	135 Mb	430 Mb	500 Mb
Available tools	Mutant banks, transgenesis, CRISPR, transcriptomics ...	Mutant banks, transgenesis, cultivated varieties	Possible transformation, genomics, transcriptomics, few genetic resources
Main applications	Fundamental biology, signaling, stress ...	Agriculture, water stress, biofortification	Lignification, biomass, bioenergy, ecophysiology
Specific limitations	Not representative of agronomic crops	Heavier cultivation, genetic diversity	Long cycle, classical genetics difficult

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Approaches in physiology

